

Mapping Tax Elasticities

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Preliminary and incomplete. The latest version is available [here](#).

Abstract

Existing estimates of tax elasticities rest on local, *between-income* contrasts, while the questions asked of them are broad. This paper proposes a source of variation available throughout the map of own and partner income. Because it is *within-income*—it stems from shifts to the tax liability function which, under family taxation, depend on partner income—own income can be controlled for without absorbing the identifying variation, and comprehensive heterogeneity analyses become feasible. The universe of French tax returns (2003–2023) reveals elasticities that decline with income, and, for women, participation responses concentrated in households with an unequal income split. Answers to normative and positive questions—where to tax income more/less, the effects of reforms to jointness—are sensitive to this heterogeneity.

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1 Introduction

When a picture contains many moving parts, it can help to zoom in. To measure tax responsiveness in tax data, the main problem is simultaneity: tax rates are a function of income, and the effort to earn income, a function of how much it gets taxed, which means that a naive regression of income growth on tax-rate changes does not give a causal estimate of a “tax elasticity”. In response, the literature has gone local: (i) zooming in on thresholds at which marginal tax rates rise, inferring tax elasticities from the excess density of taxpayers just below them—the idea behind bunching designs (H. J. Kleven, 2016; Saez, 2010), or (ii) zooming in on a group affected by a tax reform, approximating its counterfactual earnings growth with a similar group left unaffected—the idea behind simulated tax-reform IV designs (Gruber & Saez, 2002).

At the same time, the questions for which these estimates are needed often take the zoom out. The tax elasticity of labour supply (or, “sufficiently”, of taxable income; Feldstein, 1999) is a key input to optimal-tax models, various macroeconomic models as well as microsimulations of the effects of reforms—and it may matter which local estimate is used: an elasticity identified from a cut to top rates need not be close to any representative average. And many questions are comparative: the costs of a progressive tax schedule depend on the tax elasticities’ income gradient (Saez, 2001), and the costs of a joint tax schedule (taxing married couples’ income as one), on the ratio of spousal elasticities (Boskin & Sheshinski, 1983). The dispersion of existing estimates is large (Neisser, 2021), and it is difficult to establish how much of it reflects heterogeneity in tax responsiveness and how much the populations, margins and designs that differ across studies.

This paper proposes a source of variation suited to identifying heterogeneities across income, spousal income split and gender. Like bunching, the approach exploits discontinuities in the tax schedule, but it does so in the context of “events” that change the tax liability function—such as, under family taxation, the fiscal exit of a child from the family tax unit or marriage. It relies on the fact that, under the French system, differences in partner income make otherwise identical individuals differentially likely to switch tax bracket: two identical women with the same earnings, e.g., both in the 14% bracket, whose first-born reaches the age at which fiscal unit attachment ends, may see different tax changes—one remaining at 14%, the other, with

a somewhat higher-earning partner, pushed into the 30% bracket. The idea is to use the assumption that there is no anticipatory bunching around not-yet-effective (childless or joint) thresholds.

Two features of this variation make it interesting. (i) It comes from shifts of the bracket thresholds of otherwise unchanged tax schedules, which can be analysed one at a time—both supporting a weakly causal interpretation of the estimands (leaving the schedule itself unchanged makes monotonicity more plausible than under reform variation, and analysing one shift at a time avoids the negative weights that heterogeneous responses can generate; Graber et al., 2026). (ii) It is within- rather than between-income. When exposure to tax variation is a function of own income—as with a reform affecting a particular part of the income distribution—income controls face a tension: they must be rich enough to absorb mean reversion and secular inequality trends, but not so rich as to absorb the identifying variation.¹ When the source of variation is partner income, it becomes safe to control for income, possible to illustrate the assumptions in intuitive graphs (adapting Jakobsen and Søggaard, 2022), and feasible to run some heterogeneity analyses—in one observational setting. For such multi-dimensional heterogeneity analysis, correspondingly large data are needed. From the universe of French tax returns from 2003 to 2023—filings at the tax-unit level—we built an individual-level panel.² It is the size of the data that permits a very granular analysis: *within* narrow cells of the own and partner income distribution, we analyse every possible event-induced tax change separately and then aggregate—and do so by gender and event type.

We find elasticities of own income with respect to the household net-of-tax rate that decline with income—most steeply at low incomes—and are shifted upward for women: averaged over the French taxpayer distribution, the intensive-margin elasticity is 0.29 (0.33 for women, 0.26 for men), and the corresponding participation response is 0.39 (0.34 for women, 0.43 for men). For women the two margins point in opposite directions across the household’s income split. Among the lowest earners, their participation responds most strongly when the partner

¹Gruber and Saez (2002) and Weber (2014) develop approaches to address mean reversion and shifts in the income distribution, with related approaches in Blomquist and Selin (2010), H. J. Kleven and Schultz (2014) and Miao et al. (2025); Kumar and Liang (2020) additionally allow responses to vary across individuals.

²Following individuals through events such as marriage requires linking them across tax units, which individual-level identifiers in the data permit.

earns considerably more (0.52, against 0.14 when the partner also earns little), whereas their intensive-margin response is largest where both partners earn little (0.71). Men’s participation responses are likewise largest at low own incomes.

Replacing constant elasticities with profiles of the estimated shape changes the conclusions of optimal-tax models and reform evaluations. In illustrative calibrations that mimic our estimated profiles (Section 5), feeding the income profile into the sufficient-statistics formula for optimal taxation (Saez, 2001), in the implementation of Brewer et al. (2010), shifts the schedule’s progressivity up the income distribution and flattens it at low incomes, relative to the constant-elasticity benchmark; and replacing the constant per-spouse elasticities in the evaluation of reforms towards more individual taxation (Bierbrauer et al., 2024) changes the predicted employment effects and raises the assessed welfare gains.

Holding constant what is not constant across studies—setting, design, and outcome—and extending the evidence to lower incomes can be seen as the paper’s main contribution to the literature documenting heterogeneity in tax responsiveness by income. Figure 1 collects the quasi-experimental estimates of work income responses published in top field or general interest journals. It shows that the evidence comes from narrow slices of the distribution or gives only a single average over a wide range, and that the top three deciles have been more studied than the rest of the distribution.

For the literature on the taxation of couples, the relevant feature of our variation is that it also spans partner income. Normative work gives opposing prescriptions on optimal rates depending on the assumed difference between primary- and secondary-earner elasticities (Boskin & Sheshinski, 1983; Golosov & Krasikov, 2025).³ Positive work, studying actual or comparable switches between joint and individual taxation, finds that joint taxation lowers the employment and earnings of secondary earners but not of primary earners, typically men (Herold et al., 2025; Herold & Wallossek, 2024; Isaac, 2023; Kalíšková, 2014; LaLumia, 2008; Selin, 2014)—in line with what the heterogeneity we map by own and partner income would predict.

Other margins of heterogeneity to which we contribute evidence are the outcome, gender, and the time horizon. We further estimate responses of earnings inclusive of self-employment

³With a common elasticity, optimal marginal rates on secondary earners are on average *higher* than on primary earners (Golosov & Krasikov, 2025); with the secondary earner assumed more responsive, the inverse-elasticity argument prescribes the opposite (Boskin & Sheshinski, 1983).

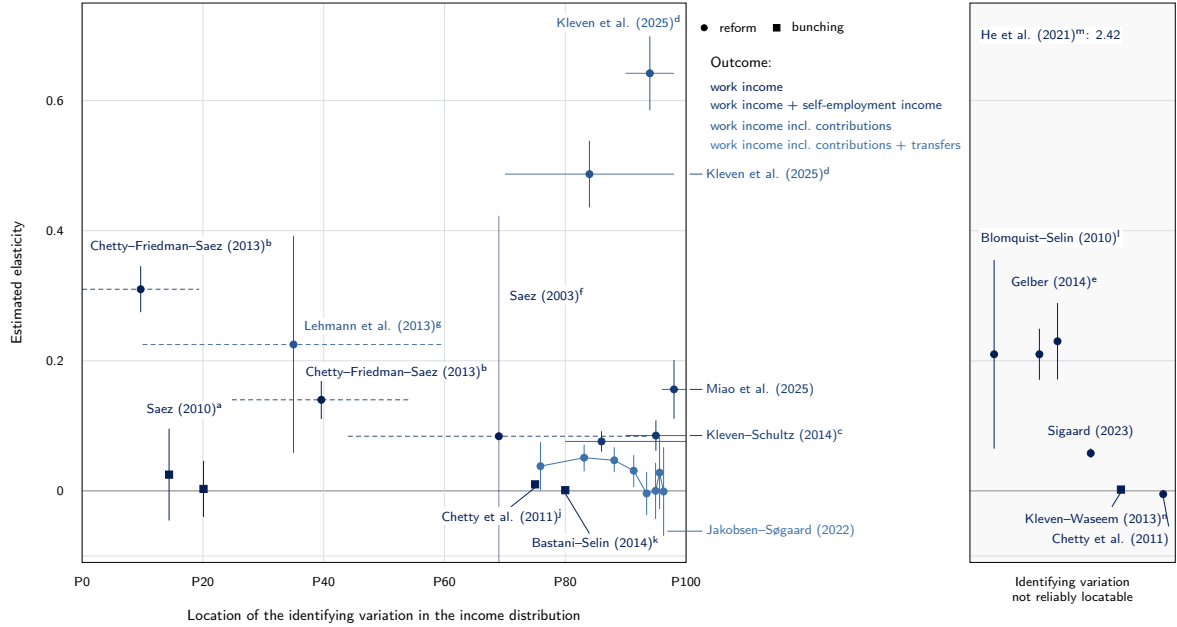


Figure 1: Published estimates of work income responses

Notes: The figure shows quasi-experimental estimates from studies published in top field or general interest journals. Colours denote the outcome (in-figure key); circles denote reform designs and squares bunching designs. Work income incl. contributions is measured gross of employer contributions (payroll contributions in Lehmann et al., employer-administered pension contributions in the Danish studies). Point markers sit at kinks or local-regression centres; horizontal segments give the income range from which the identifying variation comes (solid: reported by the study in percentile terms; dashed: computed by us). Estimates without a locatable range appear in the right panel, where horizontal position is uninformative. Vertical lines are 95% confidence intervals where an elasticity standard error is reported. Percentile ranks refer to each study's ranking universe. Papers estimating taxable-income elasticities for populations broader than wage earners^h or Frisch elasticitiesⁱ are not included. Exact populations, horizons, estimands, treated ranges and mapping grades are reported in Appendix Table H.1.

^a One-child (left) and two-plus-children (right) EITC kinks, wage earners only. ^b EITC phase-in (left) and phase-out (right) regions. ^c Top-20 and nested top-10 percent samples. ^d All treated job switchers and, above, the top decile; long-run elasticities, whereas the other reform estimates are one-to-four-year responses. ^e Husbands (left) and wives (right), both uncompensated. ^f Confidence interval truncated below; the segment covers the bracket-creep kinks holding roughly nine-tenths of treated taxpayers (full support about P21–P98). ^g Approximate: the treated interval of 0.3 to 2.1 minimum wages, which covers half of French employees. ^h Feldstein (1995) and Gruber and Saez (2002); surveyed in Saez et al. (2012). ⁱ From temporary tax holidays: 0.41 in Iceland (Sigurdsson, 2025), 0.025 in Switzerland (Martínez et al., 2021). ^j Top-kink bunching in taxable income among wage earners; no elasticity standard error is reported. The right-panel entry is the same study's schedule-wide wage-earnings reform estimate. ^k First central-government kink, taxable labour income; the paper reports 0.00 (rounded), without a standard error. ^l Taxable labour income, uncompensated; the marker is the male estimate, 0.21 (0.07); the female estimate, 1.44 (0.83), exceeds the axis and its confidence interval does not exclude zero. ^m Taxable wage income; the reform estimate, 2.42 (0.19), exceeds the axis; the same study's bunching estimates on the same schedule are 0.09–0.41. ⁿ Observed (friction-attenuated) elasticity of taxable income at the wage-earner notches, derived as the paper instructs: 0.000–0.004 across the four notches, no standard error applicable. The friction-corrected structural bounds are 0.001–0.074.

income, which closely track the work-income responses—we do not find the amplification with broader income concepts documented elsewhere (H. J. Kleven & Schultz, 2014; Sigaard, 2023); responses of deductions are under estimation.⁴ We find weaker gender differences than the literature, in which married women and single mothers are above-average elastic in labour supply (Blomquist & Selin, 2010; Eissa, 1996; Eissa & Liebman, 1996), and attribute this to the different life-cycle stages our estimates cover. With the time horizon recently highlighted as central to the gap between micro and macro estimates (H. Kleven et al., 2025), our event-time profiles separate immediate from persistent responses at horizons of one to four years.

The rest of the paper is structured as follows. Section 2 describes the French tax system and the construction of the individual-level panel. Section 3 develops the empirical framework and Section 4 presents the results. Section 5 examines how accounting for heterogeneity affects optimal tax schedules and the evaluation of reforms towards more individual taxation. Section 6 concludes.

2 Background

2.1 Institutional setting

The variation explored in this paper stems from two features of the French tax system: (i) the dependence of the tax liability on household composition through the *family quotient*, and (ii) the step-wise shape of the marginal tax schedule.

What makes the French tax system one of “family taxation” is that the tax burden (equation (1)) is a function of the family quotient, which increases with the number of family members: the quotient is 1 for singles, 2 for childless married (or civil-union) couples and 3 for a married couple with two children, increasing for each child up to a cap.⁵⁶

⁴Our variation does not identify responses of taxable or capital income—the object of a large literature (e.g., Feldstein, 1995; Gruber & Saez, 2002; Neisser, 2021; Saez et al., 2012).

⁵The tax advantage associated with children is capped, such that the benefit from an additional child does not exceed a predetermined threshold.

⁶Other factors also influence the family quotient. For example, it increases for single parents and parents of disabled children. We do not consider these cases in our analysis.

The tax liability is defined (i) under individual filing and (ii) under joint filing as

$$(1) \quad (i) \ T_t(z_{it}), \quad (ii) \ T_t^H(z_{it} + z_{pt}; Q_{it}) = Q_{it} T_t\left(\frac{z_{it} + z_{pt}}{Q_{it}}\right),$$

where T_t denotes the year- t French income tax formula for a single individual, z_{it} and z_{pt} are own and partner taxable income, and Q_{it} is the family quotient.

The resulting marginal tax rates for different family compositions are depicted in Figure 3, Panel (a). Panel (b) illustrates the resulting income tax liabilities, accounting for both the family quotient and the cap on child-related tax advantages.

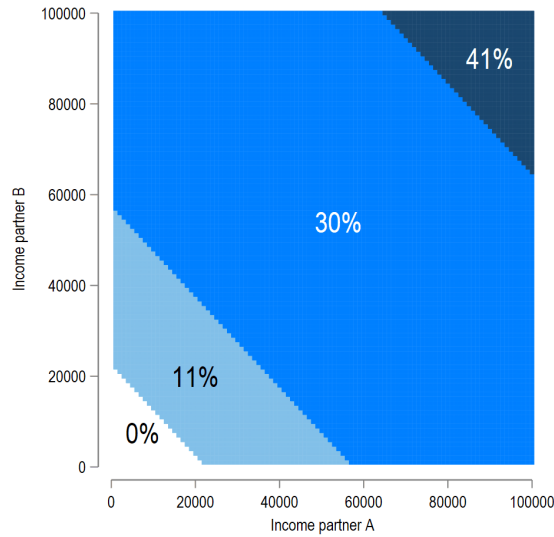


Figure 2: Marginal tax rate as a function of the income of both spouses

Notes: The heatmap illustrates how the marginal tax rate depends on the income of both partners, under the tax system of 2023. Conditional on the income of one partner, the marginal tax rate increases with the income of the other partner.

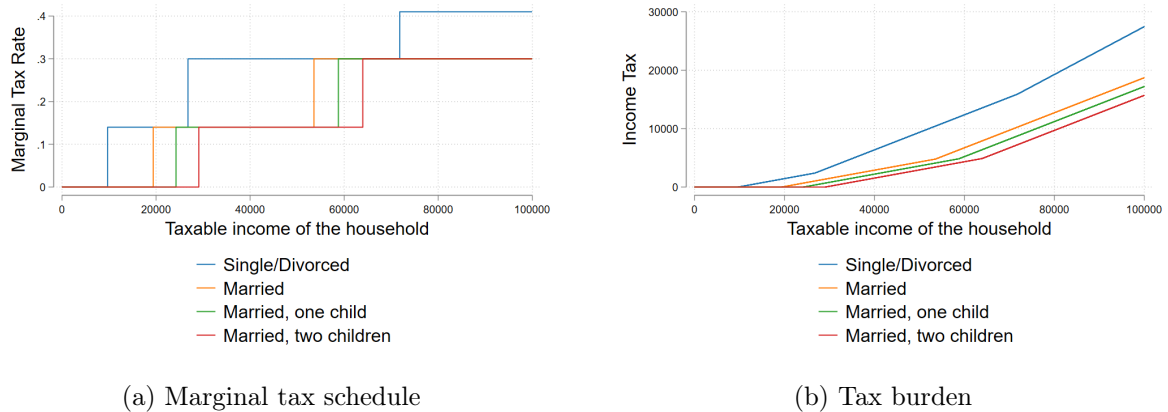


Figure 3: Marginal tax schedule and tax burden by household composition

Notes: Panel (a) displays marginal tax rates for different household compositions in 2015. Panel (b) displays the tax burden for different family situations in 2015. Source: Own calculations based on OpenFisca France.

Under family taxation, unlike under individual taxation, family events change the tax function that applies to an individual’s income—the variation studied subsequently. The fiscal exit of a dependent child lowers the family quotient and leaves the tax unit otherwise unchanged; marriage merges two tax units into one, pooling the spouses’ incomes. Table 1 illustrates the two events.

Table 1: Family events and the tax liability function

Event	Before the event	After the event
Child’s fiscal exit: one dependent child	couple + child, $Q = 2.5$	couple, $Q = 2$
two dependent children	couple + children, $Q = 3$	couple + child, $Q = 2.5$
Marriage (childless couple)	two single units, $Q = 1$ each	one joint unit, $Q = 2$

Notes: Example configurations of the two events; Q denotes the family quotient of the tax unit. Households with more dependent children see the analogous quotient change, subject to the cap on the child advantage; the marriage row shows two previously single, childless individuals merging into one jointly taxed unit.

2.2 Data

We use the French administrative tax panel known as POTE.⁷ The dataset covers all income tax returns filed in France between 2003 and 2023—all individuals required to file a French income declaration. POTE includes all variables submitted in the tax return (Form

⁷POTE—Panel On The Taxation of Households, INSEE–DGFIP. Data accessed via CASD (<https://www.casd.eu/>).

No. 2042), intermediate calculations for determining the final tax liability, and all information shown to the household on its official tax notice. It also includes detailed tax-relevant characteristics—the number of dependent children, marital status and filing status, gender, and the municipality—relevant to the subsequent analysis.

Tax returns are filed at the tax-unit level (the *foyer fiscal*), but the POTE database has the important advantage of including individual-level identifiers. It permits the (to our knowledge) first construction of a longitudinal individual-level panel, essential to our design: it lets us link individuals across tax units and follow their incomes before and after family events that change the tax liability function, such as marriage.

The result is a panel of approximately 50 million individuals per year before cleaning; the cleaned year files used in the analysis contain between 39.1 million (2003) and 42.9 million (2023) individuals. In 2023, 62 percent of men and 57 percent of women declare positive work income, and 50 and 44 percent, respectively, are married or in a civil union. From this panel we draw the two event samples: individuals experiencing the fiscal exit of their first child, and individuals marrying. Table 2 describes the two samples.

Table 2: The two event samples

	Child's fiscal exit		Marriage	
	Women	Men	Women	Men
Individuals	1,884,448	1,758,056	813,974	661,357
Annual event cohort size (mean)	96,707	96,804	191,309 ^a	195,152 ^a
Age at $r = 0$ (mean)	51.12	53.39	34.08	35.92
Partner's age at $r = 0$ (mean)	53.42	51.11	36.26	33.86
Married at $r = 0$ (share)	0.98	0.98	0.56	0.55
Dependent children at $r = -1$ (mean)	1.92	1.93	0.32	0.38
Own work income (€, mean)	18,126	27,021	18,339	22,316
Partner's income (€, mean)	42,481	21,632	24,107	18,336
Positive work income (share)	0.79	0.85	0.92	0.93
Predicted into a rate change (share)	0.30	0.32	0.27	0.23
Mean $ \text{predicted } \Delta \ln(1 - \text{MTR}) $ if treated	0.12	0.12	0.11	0.11

Notes: Descriptive statistics for the two event samples, measured in the pre-event year ($r = -1$) unless otherwise indicated, over the analysis support (own work income below €70,000). The child's fiscal exit sample is restricted to individuals whose filing status is unchanged at the event; the marriage sample comprises first marriages. Partner's income sums work, self-employment, and pension income. The mean absolute predicted change in the log net-of-tax rate is calculated among treated individuals. Counts respect the data provider's disclosure rules. ^a Annual event cohort size for the marriage sample is calculated using pre-COVID years only.

3 Empirical framework

This section sets out the model and the identification problem (Section 3.1), introduces the reform-based and event-based designs (Section 3.2), defines the cells within which comparisons are made, states the identifying assumptions, and establishes the causal interpretation of the estimands (Section 3.3), and describes aggregation, estimation and inference (Sections 3.4 and 3.5).

3.1 The model and the identification problem

We model a jointly taxed couple, extending the standard one-person model (Jakobsen & Sogaard, 2022; Saez et al., 2012) to a unitary household maximising a quasi-linear and iso-elastic utility function, with the costs of earning income additively separable across spouses.

The household solves

$$\max_{z_{it}, z_{pt}} c_t - \sum_{s \in \{i, p\}} \frac{n_{st}}{1 + \frac{1}{\epsilon_s}} \left(\frac{z_{st}}{n_{st}} \right)^{1 + \frac{1}{\epsilon_s}} \quad \text{s.t.} \quad c_t \leq z_{it} + z_{pt} - T_t^H(z_{it} + z_{pt}),$$

where z denotes income, c household consumption, and T_t^H the household tax function of Section 2.1.⁸ The cost of earning income is governed by two individual parameters: the elasticity ϵ_i and potential income n_{it} .⁹ Quasi-linearity rules out income effects, so compensated and uncompensated elasticities coincide; income effects are added in Appendix C.

Because the couple is taxed jointly on the sum of spousal incomes, the marginal net-of-tax rate is common to the two spouses, $\tau_{it}^H \equiv 1 - T_t^{H'}(z_{it} + z_{pt})$. With the separable costs (Appendix A relaxes this), the household problem separates into two one-person problems; each spouse's first-order condition, evaluated at the household rate, yields the income-supply function $z_{it} = n_{it} (\tau_{it}^H)^{\epsilon_i}$.¹⁰ Taking logs and first differences gives the estimating equation

$$(2) \quad \Delta \ln z_{it} = \epsilon_i \Delta \ln \tau_{it}^H + \Delta \ln n_{it}.$$

⁸Individual filing is the special case with a quotient of one; the marriage design, which switches the filing regime, is treated in Appendix A.

⁹To be interpreted as the income i would earn when the marginal tax rate is zero (Saez et al., 2012).

¹⁰This is the formal counterpart of the standard practice of pairing the one-person model with the marginal rates a household tax system produces (Bick & Fuchs-Schündeln, 2018; Gruber & Saez, 2002).

The parameter of interest is ϵ_i : the elasticity of own income with respect to the marginal net-of-tax rate the individual faces.¹¹ The error term, $\Delta \ln n_{it}$, varies across individuals and years—transitory shocks, life-cycle and secular growth—and, at a family event, may respond to the event itself; the design does not assume it exogenous, as Section 3.3 will make clear. Elasticities are individual throughout; the objects of interest are averages of ϵ_i over subpopulations defined by own and partner income (characterised in Section 3.4).

A naive estimation of equation (2) would not produce causal estimates of ϵ_i . Because the tax schedule depends on income, the marginal net-of-tax rate is a deterministic function of the dependent variable, which means that any change in potential income changes realised income, and with it T_t^H —the regressor of equation (2) is thus correlated with its error term by construction.

The literature handles this endogeneity in two ways: (i) bunching designs, which impose enough structure to infer the elasticity from excess mass at kinks of the schedule (H. J. Kleven, 2016; Saez, 2010); and (ii) IV designs that draw arguably exogenous variation from tax reforms—the approach we build on, and against which we compare our design in the next subsection.

3.2 Our design relative to the standard reform design

Standard, reform-based instruments. The reform-based instrument of Gruber and Saez (2002) applies the post- and pre-reform schedules to the same pre-reform income, predicting each individual’s rate change from the legislated change alone,¹²

$$\Delta \ln \tau_{it}^{\text{reform}} = \ln(1 - T_t^{\text{post}}(z_{i,t-k})) - \ln(1 - T_t^{\text{pre}}(z_{i,t-k})).$$

Validity requires

$$\text{cov}(\Delta \ln \tau_{it}^{\text{reform}}, \Delta \ln n_{it} \mid z_{i,t-k}) = 0 :$$

¹¹Appendix A relaxes separability: equation (2) is unchanged, with ϵ_i read as the elasticity of own income with respect to the household rate, $\epsilon_i^H \equiv \epsilon_{ii} + \epsilon_{ip}$ —the sum of its responses to the own and to the partner’s rate, each a jointly re-optimised household response. The appendix shows how the two events identify the components separately.

¹²See Saez et al. (2012) for an overview of the literature.

conditional on lagged income, potential-income growth must be unrelated to the predicted rate change. The assumption is demanding because exposure to reforms is usually *between*-income: the instrument is a function of lagged income, and potential-income growth is related to the level of income through mean reversion and differential secular income trends. The literature mitigates these concerns with richer income controls (Gruber & Saez, 2002; H. J. Kleven & Schultz, 2014), longer instrument lags (Weber, 2014), and differencing against placebo periods or cohorts with graphical validation (Graber et al., 2026; Jakobsen & Sogaard, 2022).

A further concern, developed in Appendix G, is that, with income effects or an extensive margin and heterogeneous elasticities, reform variation can move different individuals' realised marginal rates in opposite directions, in which case the IV estimand no longer aggregates behavioural responses with convex weights.

The proposed, event-based instrument. Instead of using reform variation, we exploit the fact that family events change the tax function that applies to an individual's income. Our main design uses the fiscal exit of a dependent child, which lowers the family quotient and leaves the tax unit otherwise unchanged (Section 2.1); marriage, the second event we use, instead merges two tax units. We index the two schedules by $g \in \{0, 1\}$: the individual is taxed under the pre-event schedule ($g = 0$) or the post-event one ($g = 1$). Fiscal attachment of a child ends automatically at age 18 and can be extended to age 21 (25 if in education). Dates are expressed in event time, $r \equiv t - m_i$, where m_i is the calendar year of the event—for the fiscal exit, the year in which fiscal attachment of the first-born ends—and the schedule indicator in event time is $g_{ir} \equiv \mathbb{1}[r \geq 0]$: every individual in our designs is taxed under the pre-event schedule for $r < 0$ and under the post-event schedule from $r = 0$ onwards.¹³ The design does not require the event to be exogenous: in the comparisons of Section 3.3, made within cells of individuals alike in pre-event own income, partner income, event year and event type, treated and control individuals all experience the event, and identification requires only that its direct, non-tax effect on income growth be common across doses (the predicted rate changes defined below) within a cell (Assumption 1), and that the timing of the event not sort on the tax consequence—checked with density tests and pre-trends (Appendix D).

¹³In its simplest form, the design is therefore a two-period comparison—the pre-event year $r = -1$ against a post-event year; the event-time profiles of Section 4 extend it horizon by horizon.

Per-part income under schedule g , the income to which the bracket schedule applies, is total tax-unit income divided by the quotient, $b_{ir}(g)$ —for an individual in a couple filing jointly, $b_{ir}(g) = (z_{ir} + z_{pr})/Q_g$; for an unmarried individual, $b_{ir}(0) = z_{ir}$. The instrument applies the pre-event year’s bracket schedule to pre-event ($r = -1$) per-part incomes under $g = 0$ and $g = 1$:

$$(3) \quad Z_{ir} \equiv g_{ir} \times \underbrace{\left[\ln(1 - T'_{-1}(b_{i,-1}(1))) - \ln(1 - T'_{-1}(b_{i,-1}(0))) \right]}_{\equiv Z_i, \text{ the dose}},$$

where schedules are dated in event time as well, T_{-1} being the schedule of the individual’s pre-event calendar year.¹⁴ At the child’s fiscal exit the tax unit is unchanged and the quotient falls—from 2.5 to 2 with one dependent child, from 3 to 2.5 with two—so per-part income weakly rises at every level of household income and the event predicts weakly higher marginal rates for every individual, $Z_i \leq 0$.¹⁵

This instrument definition leaves *within-income* variation: conditional on own income and the pre-event bracket, whether or not an individual is pushed over a bracket threshold is a function of partner income (Figure 2: at fixed own income, the predicted rate change varies with partner income).¹⁶

3.3 Identification and an illustration

Given the two schedules, we let $Y_{ir}(g)$ denote potential log income at event time r under schedule $g \in \{0, 1\}$, and $\text{NTR}_{ir}(g)$ the corresponding potential log net-of-tax rate; the post-event schedule applies from $r = 0$ onwards (g_{ir} , Section 3.2). $Y_{ir}(0)$ is not a no-event counterfactual: the event occurs under both potential outcomes; the schedule alone is varied. The potential outcomes map to observed outcomes through

¹⁴ Z_{ir} is the excluded instrument of Graber et al. (2026)—the interaction of a post-reform indicator with the simulated rate change—written in event time.

¹⁵ At marriage, two tax units with quotient 1 merge into a single unit with quotient 2; predicted increases and decreases coexist, and we restrict the analysis to observations with predicted increases, so that $Z_i \leq 0$ holds in every design we use.

¹⁶ Jakobsen and Sogaard (2022) develop the associated identifying assumption of constant trend differentials and its graphical validation.

$$\begin{aligned}
(4) \quad Y_{ir} &= Y_{ir}(0) + \mathbb{1}[Z_{ir} \neq 0] [Y_{ir}(1) - Y_{ir}(0)], \\
\text{NTR}_{ir} &= \text{NTR}_{ir}(0) + \mathbb{1}[Z_{ir} \neq 0] [\text{NTR}_{ir}(1) - \text{NTR}_{ir}(0)].
\end{aligned}$$

For zero-dose individuals the two schedules prescribe the same marginal rate at their pre-event income—the schedule change is a treatment only where $Z_i \neq 0$ —so the mapping assigns the predicted-control group its untreated outcomes at every horizon; the group’s realised rate changes, its first stage, provide a direct check (Appendix D).

In line with the model of Section 3.1, we assume potential log income is linear in the potential rate, with the slope invariant across the two schedules (as in Graber et al., 2026; Saez et al., 2012):

$$(5) \quad Y_{ir}(\text{NTR}, g) = \epsilon_i \times \text{NTR} + \ln n_{ir}(g),$$

where ϵ_i is the individual elasticity of Section 3.1 and $n_{ir}(g)$ is potential income (the zero-rate income of the model), whose schedule argument collects every channel from the schedule change to income other than the marginal rate—the argument Assumption 2 below restricts.¹⁷

For both the identification argument and the estimation that follows, we compare at a disaggregated level, for two reasons—the credibility of the identifying assumptions and the heterogeneity of the elasticities. Comparisons are made within cells of individuals alike in the determinants of income growth, and each dose δ in the support of Z is compared separately to the zero-dose group of its cell. The identifying assumptions then need to hold only among near-identical individuals, where they are easiest to defend and to test; and with heterogeneous elasticities, pooling doses can place negative weights on individual responses (Callaway et al., 2026; Graber et al., 2026), whereas dose-by-dose contrasts aggregate them transparently.¹⁸ Formally, cells fix own income within narrow bins, the event year and the event type exactly, and partner income within an interval. Let $X_i \equiv (z_{i,-1}, z_{p,-1}, m_i, e_i)$ collect these predeter-

¹⁷With non-separable costs, the household model of Appendix A delivers the same form locally, with ϵ_i read as the elasticity of own income with respect to the household rate, $\epsilon_i^H = \epsilon_{ii} + \epsilon_{ip}$.

¹⁸Conversely, were elasticities homogeneous within cells, the dose separation and the monotonicity condition below would be dispensable—the pooled within-cell contrast identifies the common ϵ . Even then the cells would remain necessary, because treated and controls differ unconditionally in own income, event year and event type, the dimensions along which untreated growth varies.

mined variables, with e_i the event type (the pre-event constellation: the number of dependent children and single or joint filing); let the cells $\mathcal{I}_c$ be the elements of the resulting partition of the support of X , and write $c(X_i)$ for the cell of individual i .¹⁹ Throughout, expectations subscripted by c , $\mathbb{E}_c[\cdot]$, are taken over the individuals of cell $\mathcal{I}_c$.

Within cells, we impose three assumptions:

Assumption 1 (Conditional common trends). *For every cell, horizon r , and dose δ in the within-cell support of Z —writing $\Delta_r V \equiv V_r - V_{-1}$ for growth relative to the pre-event year—*

$$\begin{aligned}\mathbb{E}_c[\Delta_r Y(0) \mid Z_i = \delta] &= \mathbb{E}_c[\Delta_r Y(0) \mid Z_i = 0], \\ \mathbb{E}_c[\Delta_r \text{NTR}(0) \mid Z_i = \delta] &= \mathbb{E}_c[\Delta_r \text{NTR}(0) \mid Z_i = 0].\end{aligned}$$

Assumption 1 holds whenever untreated growth does not vary within cells at all, $\mathbb{E}[\Delta_r Y(0) \mid X_i = x] = f_y^r(c(x))$ for all x , and likewise for $\Delta_r \text{NTR}(0)$: since own income, the event year, and the event type are fixed by the cell, this requires that the exact position of partner income within the cell’s partner-income band does not predict untreated growth. By the potential income function, $\Delta_r Y(0) = \epsilon_i \Delta_r \text{NTR}(0) + \Delta_r \ln n(0)$, so the restriction concerns the untreated evolution of net-of-tax rates and of potential income—the latter including the direct effects of the event, which occur under both schedules and difference out between the two groups. Although stated conditional on the constructed dose Z_i , the assumption restricts only predetermined variation: within a cell, Z_i is a deterministic function of the position of the pre-event incomes within the cell’s bands—predominantly of partner income, own income moving only within its narrow bin—so conditioning on $Z_i = \delta$ partitions the cell by predetermined income positions, and Assumption 1 is a restriction on untreated growth across these predetermined groups. Sorting of the event’s timing on the tax consequence—say, delaying a child’s fiscal exit by extending attachment when the predicted bracket change is costly—would violate this restriction within cells, which fix the event year; the density tests of Appendix D target such sorting directly.

How can Assumption 1 be assessed, and what does it mean economically? Figures 4 and 5 explain with an example: individuals with pre-event own income between €34,000 and €36,000,

¹⁹Given X_i , both the pre-event tax bracket and the instrument of equation (3) are determined; within a cell, variation in Z_i comes almost entirely from partner income, own income moving only within its narrow bin.

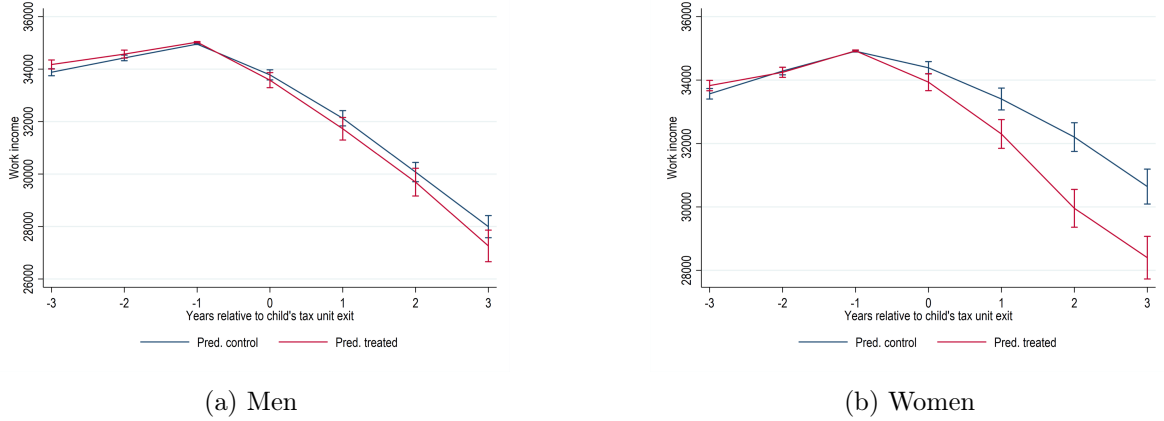


Figure 4: Work income around the first-born's fiscal exit, by predicted treatment status

Notes: Raw means for the example, without matching or covariate adjustment, so the figure shows exactly the comparison the design uses. Work income is measured relative to the pre-event year; the vertical line marks the event. The series distinguish individuals predicted to change tax bracket at the event ($Z_i = \delta < 0$) from those predicted not to ($Z_i = 0$). Table 3 describes the sample.

at a pre-event marginal rate of 14%, in 2.5-part tax units whose first-born exits at $r = 0$. Individuals whose partner earns enough are predicted into the 30% bracket once the quotient falls to 2 ($Z_i = \delta = \ln(0.70/0.86) \approx -0.21$), and the rest are predicted to remain in the same bracket ($Z_i = 0$). Figure 4 plots the income paths of the two groups; their alignment before the event is the assumption's testable implication. Figure 5 shows the split of the sample by partner income into a *treatment region*, where the exit is predicted to raise the marginal rate, and a *validation region*, where it is not. Economically, the assumption requires partner income to have no channel to income growth of its own within the example's partner-income range; the validation region checks this directly, since there partner income varies with no accompanying rate change: a flat profile of income growth in partner income across the validation region is the testable implication. Section 4 reports the same checks aggregated over all cells of the design: pooled pre-event contrasts (Figure 6) and the pooled analogue of Figure 5 (Figure 7).²⁰

Assumption 2 (Exclusion restriction). $n_{ir}(0) = n_{ir}(1)$ with probability one.

Potential income—including any direct effect of the event, through time use, home production, or transfers—is unaffected by the schedule change: the event occurs under both schedules and

²⁰Narrower partner-income bands weaken the requirement, and the matching implementation of Section 3.5 additionally aligns pre-event income paths within cells.

Table 3: The example cell: characteristics by predicted treatment status

	Women		Men	
	Control	Treated	Control	Treated
N	2,027	1,755	4,684	2,576
Work income (€)	34,909	34,922	34,952	35,027
Partner's work income (€)	23,109	30,372	21,297	28,943
Age	53.2	53.8	54.0	54.7
Partner's age	54.2	54.9	52.1	53.1
Marginal rate at $r = 1$, predicted	14%	30%	14%	30%
realised (mean)	16%	25%	16%	24%

Notes: The two groups are balanced on own income and age, and differ in partner income, which determines whether the fall in the family quotient pushes per-part income across the threshold of the post-event schedule. All rows are measured in the pre-event year, except the rate rows, which refer to $r = 1$.

is left unrestricted, while the schedule change affects income only through the marginal net-of-tax rate, the potential-outcomes form of the no-income-effects property of the quasi-linear model of Section 3.1.²¹ Under quasi-linearity the assumption holds by construction.²² With income effects, the change in tax *liability* is a second channel from the schedule change to income: Assumption 2 fails, and the estimands below recover a total response to the event-induced schedule change. Appendix C shows how to relax the assumption, and Figure C.1 there illustrates the approach.²³

Assumption 3 (Monotonicity). $\text{NTR}_{ir}(1) \leq \text{NTR}_{ir}(0)$ with probability one, for all $r \geq 0$.

The events we use predict weakly higher marginal rates for every treated individual ($Z_i \leq 0$; Section 3.2), and no individual's rate may respond to the schedule change against this prediction—rates remain free to move either way for reasons unrelated to the event. This is the standard no-defiers condition ensuring that Wald estimands reflect positively weighted averages of individual parameters (Angrist & Imbens, 1995; Imbens & Angrist, 1994). Violations are less likely when, as here, the event shifts the thresholds of the schedule rather

²¹Formally, decompose $\ln n_{ir}(g) = \mu_{ir} + \xi_{ir}(g)$ with $\xi_{ir}(0) \equiv 0$: μ_{ir} carries life-cycle and calendar-time growth and the event's direct effects—changed time use, home production, transfers—which occur under both schedules and remain unrestricted; Assumption 2 sets $\xi_{ir}(1) = 0$. Reform settings impose the analogous restriction when they assume potential income unaffected by the reform.

²²Quasi-linearity also implies that ϵ_i is the compensated elasticity, which coincides with the uncompensated one (Graber et al., 2026, Propositions 3–4).

²³The event-induced liability change scales with the amount of income shifted across the threshold and can be predicted from pre-event incomes in the same way as the rate change; Appendix C outlines how it could serve as a second, event-based instrument to identify income effects separately.

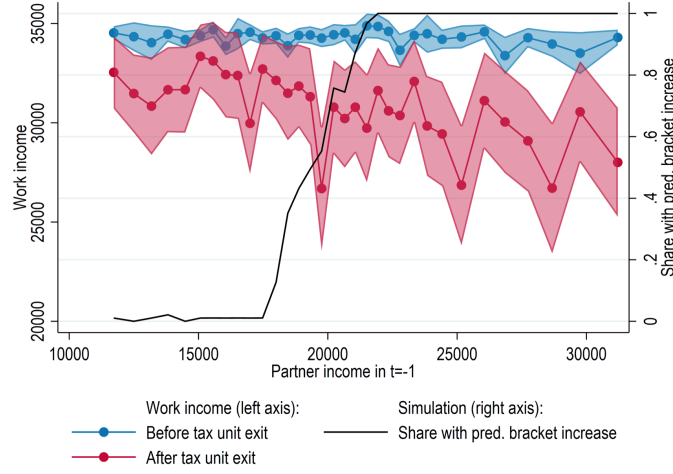


Figure 5: Identifying variation at the first-born's fiscal exit

Notes: Sample: women of the example (pre-event own income in $[\text{€}34,000, \text{€}36,000]$, pre-event marginal rate 14%, 2.5 fiscal parts, first-born exiting the tax unit at $r = 0$). The figure follows the graphical validation approach of Jakobsen and Sogaard (2022). In the validation region the event predicts no rate change, so a gradient of income growth in partner income there would indicate a direct partner-income channel, violating Assumption 1. The corresponding figure for men is Figure D.1 in Appendix D.

than its gradient; Appendix G contrasts this with reform settings—where income effects and extensive-margin responses easily move realised rates against the prediction—and illustrates the argument with a simulation.

For each cell, dose δ , and horizon, the reduced-form and first-stage contrasts compare realised income growth $\Delta y_{ir} \equiv \ln z_{ir} - \ln z_{i,-1}$ (the observed counterpart of $\Delta_r Y$) and net-of-tax rate growth $\Delta \text{NTR}_{ir} \equiv \ln \tau_{ir}^H - \ln \tau_{i,-1}^H$ across dose groups:

$$RF_c^r(\delta) \equiv \mathbb{E}_c[\Delta y_{ir} \mid Z_i = \delta] - \mathbb{E}_c[\Delta y_{ir} \mid Z_i = 0],$$

$$FS_c^r(\delta) \equiv \mathbb{E}_c[\Delta \text{NTR}_{ir} \mid Z_i = \delta] - \mathbb{E}_c[\Delta \text{NTR}_{ir} \mid Z_i = 0],$$

and our headline estimand is the cell-dose elasticity²⁴

$$\beta_c^r(\delta) \equiv \frac{RF_c^r(\delta)}{FS_c^r(\delta)}.$$

²⁴Dividing the reduced form by the predicted change instead, $RF_c^r(\delta)/\delta$, scales the response by the intended rather than the realised rate change—the policy elasticity of Hendren (2016); we use this reduced-form variant where noted.

Proposition 1 (Causal interpretation). *Under the mapping (4), the potential income function (5), and Assumptions 1–3, for every cell, dose $\delta \neq 0$, and horizon $r \geq 0$ with $FS_c^r(\delta) \neq 0$:*

$$\beta_c^r(\delta) = \mathbb{E}_c[\omega_{ir} \epsilon_i \mid Z_i = \delta], \quad \omega_{ir} \equiv \frac{\Phi_{ir}}{\mathbb{E}_c[\Phi_{ir} \mid Z_i = \delta]},$$

where $\Phi_{ir} \equiv \text{NTR}_{ir}(1) - \text{NTR}_{ir}(0)$ is the effect of the event-induced schedule change on rates, and the weights $\omega_{ir} \geq 0$ average to one: the cell-dose elasticity is a positively weighted average of individual elasticity parameters among the predicted treated.²⁵ (Proof: Appendix B.)

3.4 Aggregation

We aggregate in two steps.²⁶ Within a cell, the dose-specific elasticities are combined with frequency weights over the predicted treated,

$$\beta_c^r \equiv \sum_{\delta \in \mathcal{D}_c} w_c(\delta) \beta_c^r(\delta), \quad w_c(\delta) \equiv \Pr_c(Z_i = \delta \mid Z_i \neq 0),$$

where $\mathcal{D}_c$ denotes the within-cell support of non-zero doses. In the second step, we aggregate all cells contained in a group h of interest—own-income intervals for the income profiles, and own-by-partner-income rectangles for the heatmaps of Section 4:

$$\epsilon_h^r \equiv \sum_{c \in h} v_h(c) \beta_c^r,$$

where $v_h(c)$ is cell c 's share of the group's predicted-treated individuals. We report ϵ_h^r as an event-time profile; the headline displays of Section 4 pool horizons $r = 0$ to 3, and the horizon-by-horizon profiles are collected in Appendix E.4 (currently withheld pending a data check).

²⁵Proposition 1 is the analogue, for event-based instruments, of the characterisation in Graber et al. (2026) of when reform-based ETI estimands are causal: their two conditions—a binary instrument contrast, and income controls rich enough to reproduce the conditional mean of the instrument—appear here as a feature of the design (each estimand contrasts $\mathbb{1}\{Z_i = \delta\}$ against the zero-dose group) and as Assumption 1.

²⁶The construction is stated for the intensive margin. For the extensive margin, the outcome is the participation indicator $P_{ir} \equiv \mathbb{1}\{z_{ir} > 0\}$, in changes relative to the pre-event year, $\Delta_r P_i \equiv P_{ir} - P_{i,-1}$; the same cell-dose contrasts and aggregation apply, and the coefficient, rescaled by pre-event participation, is an elasticity of participation with respect to the household net-of-tax rate.

Because both steps use non-negative frequency weights, the summary parameters inherit the causal interpretation of Proposition 1: substituting, ϵ_h^r is a weighted average of the individual parameters ϵ_i among the predicted treated of the group, with weights proportional to ω_{ir} —the frequency weights exactly offset the within-cell-dose averaging, so ϵ_h^r is the straight average of $\omega_{ir} \epsilon_i$ over the group’s predicted treated. The aggregation weights $v_h(c)$ and $w_c(\delta)$ are known sample frequencies rather than estimator-implied—in contrast to pooled regressions across treatment intensities, whose implicit weights can turn negative under heterogeneity (Callaway et al., 2026; Graber et al., 2026)—and alternative policy-relevant weightings can be substituted without affecting the identification argument.²⁷

3.5 Estimation and inference

Estimation is by two-stage least squares within cells, separately for women and men, over an event window spanning $r = -3, -2$ and $r = 0, \dots, 3$, with $r = -1$ the base year. The comparison sample at horizon r pools the dose- δ treated with the zero-dose group, and we estimate the regression

$$(6) \quad \Delta y_{ir} = a_c^r(\delta) + \beta_c^r(\delta) \Delta \text{NTR}_{ir} + W_i' \gamma + u_{ir},$$

instrumenting the realised rate change ΔNTR_{ir} with Z_{ir} ; the controls W_i are fixed effects for own income at $r = -1, -2$ and -3 (in bins), age, marital status, the number of children at $r = -1$, and the event year. The estimation cells fix the own-income bin, the partner-income band and the pre-event bracket, and pool event years, which the event-year fixed effect absorbs; the remaining dimensions of the cells of Section 3.3 are constant within cells, and the controls bind only where estimation pools more coarsely.²⁸ Estimating the reduced form and first stage on Z_i at pre-event horizons $r < 0$ —where $Z_{ir} = 0$, so (6) itself has no first

²⁷Earlier working versions additionally used inverse-variance weights within groups; these remain non-negative and preserve the causal interpretation, but weight cells by estimation precision rather than population frequency. The headline displays of Section 4 use the frequency weights above throughout; precision-weighted variants are retained in the sensitivity material.

²⁸The reduced form $RF_c^r(\delta)$ and first stage $FS_c^r(\delta) = \Delta y_{ir}$ and ΔNTR_{ir} regressed on Z_i with the same controls—are estimated alongside; the two-stage least squares coefficient equals their ratio, and without the controls the coefficients equal the group-mean contrasts of Section 3.3 divided by the dose δ —a factor common to both that cancels in the ratio.

stage—produces the placebo contrasts implied by Assumption 1, which we report alongside the post-event estimates.

The matching implementation replaces the regression comparison within each cell with nearest neighbours (with replacement) on the Mahalanobis distance of pre-event work income, measured at $r = -3$, $r = -2$ and $r = -1$, and age, with exact matching on the event year and a caliper of 0.8; RF , FS and β are then matched differences in means and their ratio (footnote 20).

Standard errors are obtained by the delta method applied to the estimated (RF, FS) pairs, with conventional, unadjusted variances; cells are treated as independent, and the same delta-method step propagates to the aggregates β_c^r and ϵ_h^r , whose weights are sample frequencies with negligible estimation error.²⁹

4 Results

This section proceeds from validation to estimates. We first describe the comparison cells that carry the estimation and assess the identifying assumptions in the aggregate, across all cells. The main results then summarise the estimates in three ways: as population aggregates (Section 4.2.1), along the own-income distribution (Section 4.2.2), and over the joint distribution of own and partner income (Section 4.2.3). Section 4.3 examines robustness to specification choices, and Section 4.4 reports heterogeneity across household characteristics.

4.1 Comparison cells and aggregate validation

The estimation decomposes into narrowly defined comparison cells, each formed by an own-income bin, a partner-income tercile within it, a baseline bracket, and a tax-change dose. In the headline specification (work income, all event years), 281 cells reach the estimation floors for men and 217 for women, spanning nine baseline-bracket bundles and 27 (men) and 26 (women) distinct doses. Cells differ widely in size: the density of taxpayers varies along the schedule, and several bracket crossings exist only in subperiods of 2003–2023 because the schedule itself changed. Every result below is therefore a summary across cells, and we report

²⁹Dose-specific comparisons within a cell share the zero-dose control group; the induced correlation across dose-specific estimates is not yet reflected in the aggregated standard errors—a correction we will implement.

throughout which cells support each displayed estimate. Table 4 describes the cell structure by gender.

Table 4: The comparison cells

	Women	Men
Individuals	1,884,448	1,758,056
Predicted treated (share)	0.30	0.32
Grid cells (own-income bin $\times$ partner tercile)	21	21
Median cell size	81,292	64,893
Smallest cell	5,442	23,520
Largest cell	202,763	162,677
Distinct doses per cell (median)	23	29

Notes: Cell counts, size distribution and treated/control composition by gender, from the estimation-sample counts (CASD export of 21 Aug 2026), at the grid level of own-income bin (€10,000 width) $\times$ within-bin partner tercile, for the child’s fiscal exit sample with unchanged civil status. The estimation cells of Section 3.5 additionally fix the pre-event bracket and the dose; that finer split, quoted in the text, is not disclosed at cell level.

How do the tests of Assumption 1 perform in the aggregate, over all cells? Because every cell contains a binary contrast—a predicted-treated and a predicted-control group—the graphical checks introduced for the example cell (Figures 4 and 5, following Jakobsen and Sogaard, 2022) extend directly to the pooled sample. Figure 6 plots income growth relative to the pre-event year for the two groups, averaged across comparison cells. The groups move in parallel before the fiscal exit of the child, and are broadly similar in predetermined characteristics outside the assignment margin (Appendix Table E.1). After the event, income declines for men—who are on average about two years older at the event than women³⁰—while women’s income continues to grow for longer. In both cases the predicted-treated group grows less than the predicted-control group from the event year onwards: a first, unscaled indication of the response, before any adjustment for dose size or prediction strength. Figure 7 addresses the exclusion restriction: that partner income affects own income growth only through the assignment to the two groups. It examines the validation region—households experiencing the exit without a predicted bracket change—where partner income varies with no accompanying rate change. A systematic relationship between partner income and own income growth in this region would indicate a violation; the plotted growth profiles are flat through the hypothetical

³⁰Mean age at $r = -1$: 51.8 for men against 49.5 for women in the predicted-control group, and 52.8 against 50.6 among the predicted treated (Appendix Table E.1).

threshold. Sorting of the event's timing is addressed directly by pooled density tests at the threshold: the estimated discontinuities are -0.002 (s.e. 0.010) for women and -0.001 (s.e. 0.009) for men (Appendix Figure E.1).

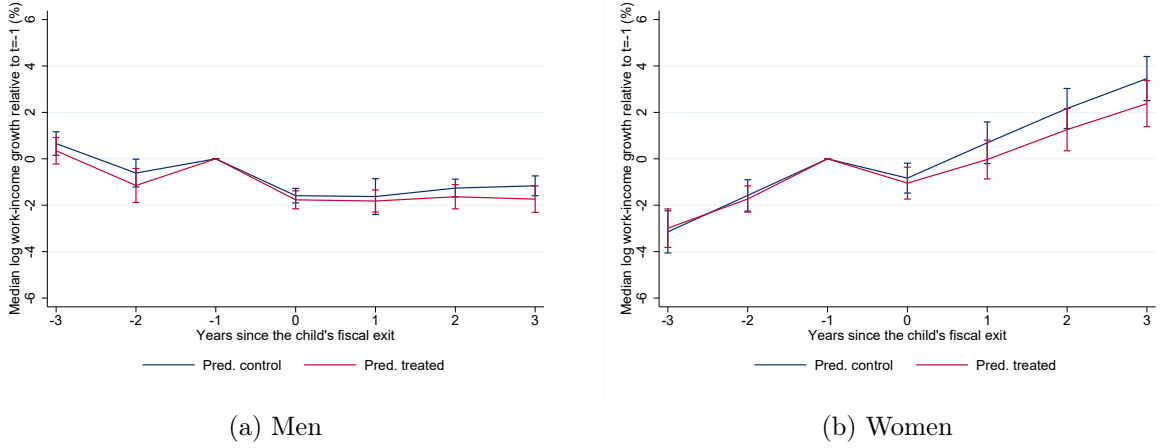


Figure 6: Pooled event study by predicted treatment status

Notes: The pooled analogue of the case-study Figure 4: median log work-income growth relative to $r = -1$ (in percent) by event year, for the predicted-control and predicted-treated groups, pooled across all comparison cells of the design interacted with event year (own-income bin $\times$ partner tercile $\times$ baseline bracket $\times$ dose; at least 30 treated individuals per dose and 300 outcomes at $r = -3$, mirroring the estimation floors). Within each comparison, a shared comparison-year mean is removed and the aggregate growth path added back, so composition across comparisons is held fixed while the scale remains interpretable; group medians are then averaged across the cell $\times$ event-year comparisons, weighted by their predicted-treated counts. Bands are 95% confidence intervals clustered by the underlying cell $\times$ event year. The outcome is mechanically zero for both arms at $r = -1$. Sample: individuals in married or civil-union tax units whose first-born exits the tax unit between 2003 and 2023, civil status unchanged at the event.

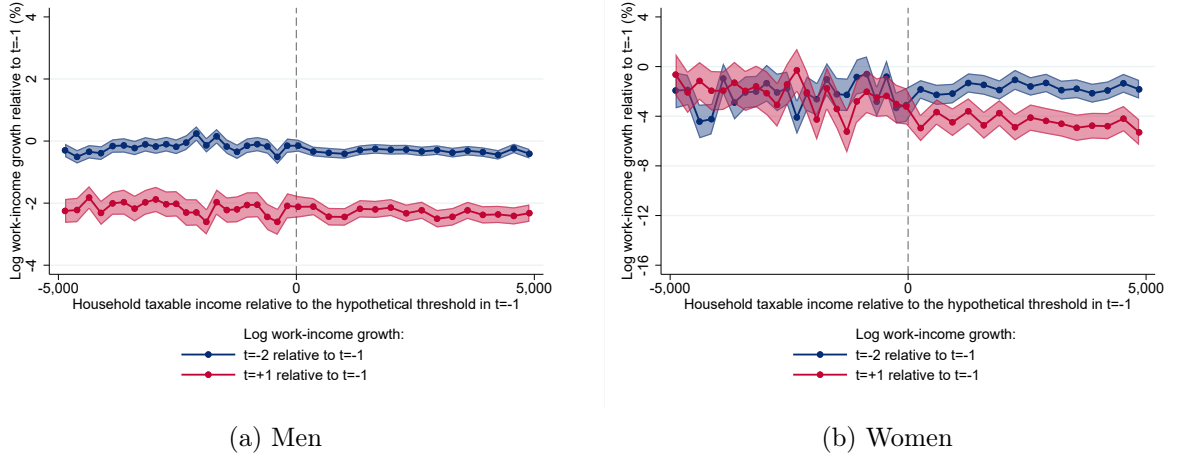


Figure 7: Identifying variation in the validation region

Notes: Log work-income growth relative to $r = -1$, at $r = -2$ (pre-event) and $r = +1$ (post-event), against the distance of household taxable income from the hypothetical post-event threshold at $r = -1$, pooled over the statutory threshold identities and displayed within $\pm\text{€}5,000$; points are raw quantile-bin means, with 95% confidence intervals clustered by person. Growth outcomes in the men’s panel are winsorised at the 5th and 95th percentiles; the women’s panel is unwinsorised. The figure is the pooled analogue of the validation-region check of Figure 5: flat profiles through zero indicate that partner income carries no income-growth gradient beyond the predicted rate change (Jakobsen & Søgaaard, 2022). Sample as in Figure 6.

4.2 Main results

The main results proceed from the most aggregated summary to progressively finer ones: population-level averages, the profile along the own-income distribution, and the map over the joint distribution of own and partner income.

4.2.1 Aggregated elasticities

What do our estimates imply for the aggregate elasticity, under the most common aggregation schemes applied in the literature? Table 5 reports four, by gender and combined; the extensive-margin counterpart is Table E.2 in Appendix E.5.

Weighted to the population of French taxpayers, the average intensive-margin elasticity is 0.293 (0.327 for women, 0.259 for men).³¹ Weighted to the event population itself—parents at the fiscal exit of their first-born, using the estimation-sample counts of the CASD result

³¹The French-taxpayer weights are constructed from the INSEE distribution of annual salary income by gender (2021), interpolated to a piecewise-linear CDF with a Pareto tail, with the non-employed entering the bottom income bin; see the table notes.

files—the average is 0.311 (0.362 for women, 0.264 for men). The two aggregates are close: reweighting from the event population to the general taxpayer distribution moves the total by less than 0.02, so the summary is not an artefact of the event population’s income composition.

The literature has also aggregated estimates with inverse-variance (efficiency) weights. That these produce the smallest average (0.212) is itself informative: the cells with high elasticities are the less precisely estimated ones, so precision weighting shifts mass towards mid-income cells where responses are modest.

The convention in the taxable-income literature, however, is to weight each individual by their contribution to the tax base. This yields an intensive-margin elasticity of 0.246 (0.019) overall—0.225 for men and 0.274 for women—below the population-weighted averages. The shortfall follows from the concentration of responses at low incomes documented in the next subsection: elasticities decline with own income for both genders, so weighting by income moves mass away from the most responsive part of the distribution.

We treat the French-taxpayer weighting as the headline aggregate; across the four schemes, the intensive-margin average lies between 0.21 and 0.31. The participation responses are of comparable size and reverse the gender ordering: weighted to French taxpayers, 0.39 overall—0.34 for women, 0.43 for men (Appendix Table E.2).

Table 5: Aggregated intensive-margin elasticities

	Men	Women	Total
Population-weighted: French taxpayers	0.259 (0.029)	0.327 (0.051)	0.293 (0.029)
Population-weighted: event population	0.264 (0.025)	0.362 (0.034)	0.311 (0.021)
Inverse-variance weighted	0.225 (0.022)	0.199 (0.022)	0.212 (0.015)
Income-weighted	0.225 (0.027)	0.274 (0.027)	0.246 (0.019)

Notes: Aggregation of tile-level estimates: cell-dose elasticities pooled within own-by-partner-income rectangles under the profile-level display conventions of Appendix E.1 (cell screens at the 0.5 tolerance, no aggregate-level screens). *French taxpayers:* prime-age taxpayers by own-income bin, from the INSEE distribution of annual salary income by gender (2021), with the non-employed entering the bottom bin; the $\leq \text{€}70,000$ support of the estimates covers 97–98% of taxpayers. *Event population:* each own-income bin $\times$ partner-tercile group is weighted by the number of individuals in its event-year ($r = 0$) estimation samples from the CASD result files; controls are shared across doses within a cell, so groups estimated under many doses are slightly overweighted. *Income-weighted:* each bin weighted by its share of aggregate earnings from the same INSEE distribution. Partner terciles are equal-weighted within bins by construction. The Total column combines genders with equal population shares (French-taxpayer and inverse-variance rows), the genders’ shares of the event-population counts (event row), and gender aggregate-income shares (income row). Standard errors in parentheses treat tile estimates as independent.

4.2.2 Maps by own income

The population averages conceal systematic variation in responsiveness, first and foremost along own income.

Figure 8 reports elasticities aggregated within own-income bins, pooled over event years $r = 0$ to 3. For women, shown in Panel (b), elasticities are largest at the bottom of the distribution, at 0.62 in the lowest bin, and lower everywhere above it: 0.10–0.16 in the €10,000–20,000 and €30,000–40,000 bins and 0.25–0.35 in the remaining bins. The €10,000–20,000 bin fails its pre-period placebo and is displayed hollow; its depression relative to the neighbouring bins partly reflects the flagged differential trend rather than behaviour. For men, shown in Panel (a), the profile declines monotonically: 0.38 in the €10,000–20,000 bin (also flagged), 0.25–0.28 through the middle of the distribution, and 0.08–0.13 in the three top bins; no estimate is displayed for the lowest bin, in which no cell passes the precision screens.

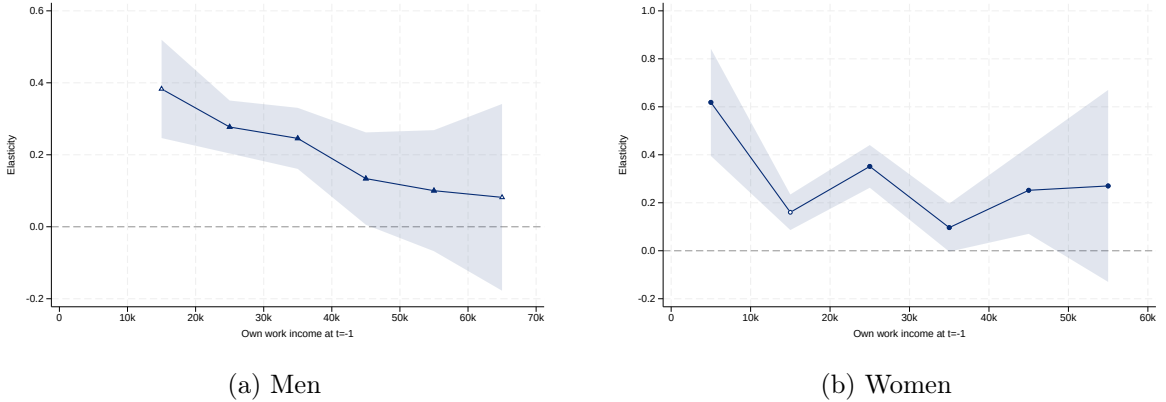


Figure 8: Intensive-margin elasticities along the own-income distribution

Notes: Elasticities decline with own income for both genders. Sample: individuals in married or civil-union tax units, civil status unchanged at the event, whose first-born exits the tax unit between 2003 and 2023 (all event years; Appendix E.9 excludes 2020–2023). The outcome is work-income growth relative to $r = -1$, pooled over event years $r = 0$ to 3; the pooling window is identical across the maps of Figure 9 and the income profiles. Construction, screens, weighting and the hollow pre-trend flags follow the display conventions of Appendix E.1, with own-income intervals as the aggregation groups. Points are placed at bin midpoints; the horizontal axis is own pre-event work income, measured at $r = -1$; shaded areas are 95% confidence intervals. The extensive-margin profiles are Figure E.2.

Are these elasticities large because of the age at which the event occurs—and is the income gradient specific to the retirement margin? The fiscal exit of a first-born typically arrives as parents approach retirement ages, and retirement is itself a margin of response. Appendix

Figure E.3 addresses both questions by repeating the analysis for individuals below age 55 at the event, and the answer differs by gender. For women, levels shift down—the bottom-bin estimate falls from 0.62 to 0.38—while the declining shape survives, indicating that their gradient is not an artefact of the retirement margin. For men, the below-55 intensive-margin profile is statistically indistinguishable from zero throughout: their intensive-margin response is concentrated among those approaching retirement ages. On the participation margin, the below-55 sample retains a large, if imprecise, response for the lowest-earning men (1.40, s.e. 0.32), while participation responses above the bottom bin fade for both genders. A complementary check exploiting a different event at another point of the life cycle—marriage—is in preparation (Appendix F).

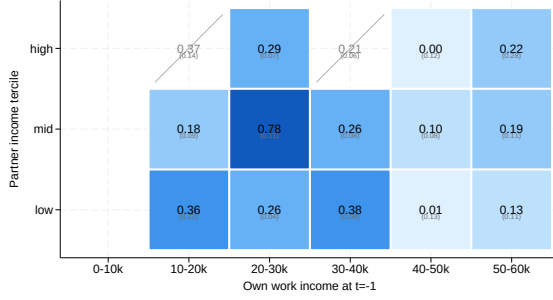
Relation to the literature. Our income-weighted intensive-margin estimate of 0.246 is within the range of quasi-experimental estimates of the elasticity of taxable income surveyed by Neisser (2021), and the decline of responsiveness with own income mirrors the concentration of participation and hours responses at low earnings documented in the labour-supply literature.

4.2.3 Maps by own and partner income

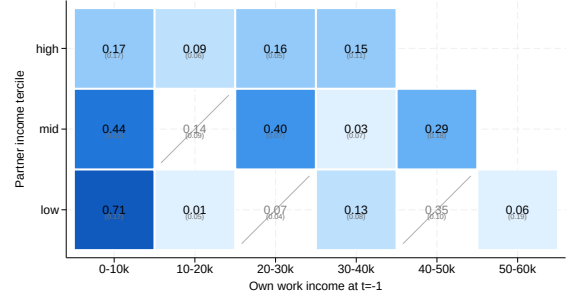
The main advantage of the design, however, is that its variation is in own *and* partner income. This permits asking which households drive the high elasticities documented at low own incomes: those insured by a high-earning partner, or those in which both partners earn little. Figure 9 disaggregates the own-income profile accordingly, reporting one estimate per own-income bin and within-bin partner-income tercile; the vertical axis orders partner income *within* each own-income bin, so the comparison across terciles holds the own-income composition fixed.

For women, the answer differs by margin. On the intensive margin (Panel b), the large bottom-bin response concentrates among households in which the partner also earns little: the low-partner tercile of the bottom bin carries an elasticity of 0.71, against 0.44 in the middle and 0.17 in the top tercile, and within every partner tercile the bottom bin carries the largest response on the women’s map. On the participation margin (Panel d) the pattern reverses: the bottom-bin response rises with partner income, from 0.14 where the partner also

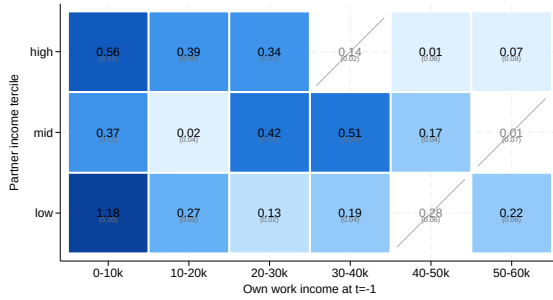
earns little to 0.52 where the partner earns considerably more. Insured women adjust on the participation margin; women in households where both earn little adjust earnings while largely remaining in work. The double-low corner is, however, where the design has least support, and the below-55 sample provides the cleaner reference for it. For men, no bottom-bin tile passes the display screens on the intensive margin (Panel a)—the €0–10,000 column is blank—and their intensive responses are spread evenly over the map, without a clear partner-income gradient; on the participation margin (Panel c), responses are sizeable across the map, with the largest estimates at low own incomes. The darkest tile of the men’s intensive map—0.78 at €20,000–30,000 with a mid-earning partner—rests on a single dominant comparison whose dose group fails a disaggregated pre-period test, and should not carry weight in the reading.



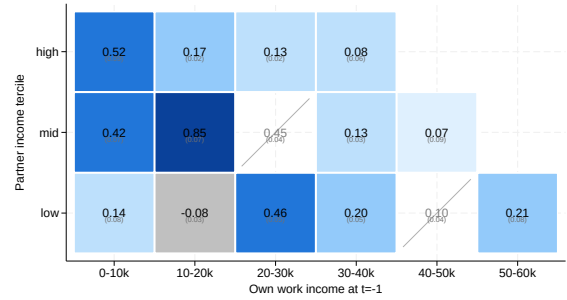
(a) Men, intensive margin



(b) Women, intensive margin



(c) Men, extensive margin



(d) Women, extensive margin

Figure 9: Elasticities by own and partner income

Notes: Within every partner tercile the bottom bin carries women's largest response, and the largest tile (0.71) is the double-low corner. Sample: individuals in married or civil-union tax units, civil status unchanged at the event, whose first-born exits the tax unit between 2003 and 2023 (all event years; Appendix E.9 excludes 2020–2023). The outcome is work-income growth relative to $r = -1$, pooled over event years $r = 0$ to 3. Each tile is one estimate—cell-dose elasticities pooled within the own-by-partner-income rectangle—with construction, screens and display rules following the conventions of Appendix E.1; the printed standard error beneath each estimate is the pooled standard error of the tile. Colour classes change at 0, 0.075, 0.15, 0.225, 0.3, 0.4, 0.55, 0.8, 1.2 and 1.7; negative estimates are grey. Tiles crossed out carry estimates whose rectangle fails its pre-event placebo contrasts (Section 3.5; reduced form pooled over $r = -3$ and -2 , $|t| \geq 1.96$); their values are printed in grey for completeness but excluded from the colour scale. Blank tiles are not estimable or fail the display screens. Both incomes are measured at $r = -1$; partner terciles are data-driven, ranked within bins by their upper boundaries, and their euro boundaries vary across own-income bins. In Panels (c) and (d) the outcome is an indicator for positive work income and the displayed estimates are participation semi-elasticities; construction and screens are otherwise identical. The complete tile-level estimates, standard errors and cell counts are reported in Appendix Tables E.3 and E.4.

Appendix E.8 shows the equivalent income profiles for alternative outcome definitions: earnings, which add self-employment income to work income, and deductions alone. The earnings profiles closely track the work-income profiles.

Is the map robust to the sample period and to the event definition itself? Appendix F will report the estimates obtained from the marriage design, and Appendix E.9 shows the maps re-estimated excluding the 2020–2023 event years, which the headline specification includes.

In sum, responsiveness declines with own income for both genders and concentrates among low earners; for women, where in the household it concentrates depends on the margin, with intensive-margin responses largest where both partners earn little and participation responses largest where the partner earns considerably more. The aggregates are robust to the weighting population (0.29–0.31 on the intensive margin, 0.33–0.39 for participation); the heterogeneity around them is large. Section 5 examines its implications for tax design.

4.3 Robustness

Table 6 summarises how the intensive-margin map of Section 4.2.3 responds to estimation variants re-estimated on the same sample generation and grid: for each variant, the correlation of tile values with the headline map, the mean absolute tile difference, and the frequency-weighted income gradient. The balanced sample reproduces the headline map almost exactly (correlations 0.95–0.96, mean absolute tile differences of at most 0.06). Nearest-neighbour matching, which replaces the regression comparison within cells (Section 3.5), reproduces the women’s map (0.70) and the negative gradient for both genders, but agrees with the men’s map only in gradient—the matched tiles are noisier. The PPML variant estimates a levels object that combines the intensive and extensive margins, so tile-level agreement with the intensive-margin map is not expected; we report it for completeness.

Table 6: Specification agreement of the heterogeneity map

	Men			Women		
	Corr.	mAD	Gradient	Corr.	mAD	Gradient
Headline	–	–	-0.07	–	–	-0.04
PPML	0.41	0.18	-0.05	-0.16	0.18	0.01
Balanced sample	0.95	0.06	-0.09	0.96	0.04	-0.05
Nearest-neighbour matching	-0.03	0.18	-0.02	0.70	0.17	-0.08

Notes: Each row re-estimates the intensive-margin map of Figure 9 under the indicated estimation variant, on the status-stable sample and headline grid, under the map-level display conventions of Appendix E.1 (pre-trend-flagged tiles included). Corr. is the correlation of tile values with the headline map on matched tiles (own-income bin $\times$ partner-tercile rank); mAD is the mean absolute tile difference on the same tiles; Gradient is the frequency-weighted slope of tile values in own income, per €10,000, also reported for the headline map. The PPML variant is estimated in levels including zeros, so its estimates combine intensive- and extensive-margin responses; the matching variant exports no first stage, so the first-stage screen does not apply to it.

4.4 Heterogeneity

The design permits heterogeneity cuts beyond own income, partner income and gender. Figure 10 reports pooled elasticities by household subgroup: the age gap between the parents, marriage versus PACS, and whether the exiting first-born is the household’s only child. Responses are larger when the age gap between the parents is four or more years (0.34 against 0.19 on the intensive margin), and larger when the exiting child is the first and only one (0.30 against 0.16)—in line with the first fiscal exit also being the last for these households. The PACS estimate rests on ten cells and is uninformative. Appendix Figure E.8 reports the by-gender breakdown.

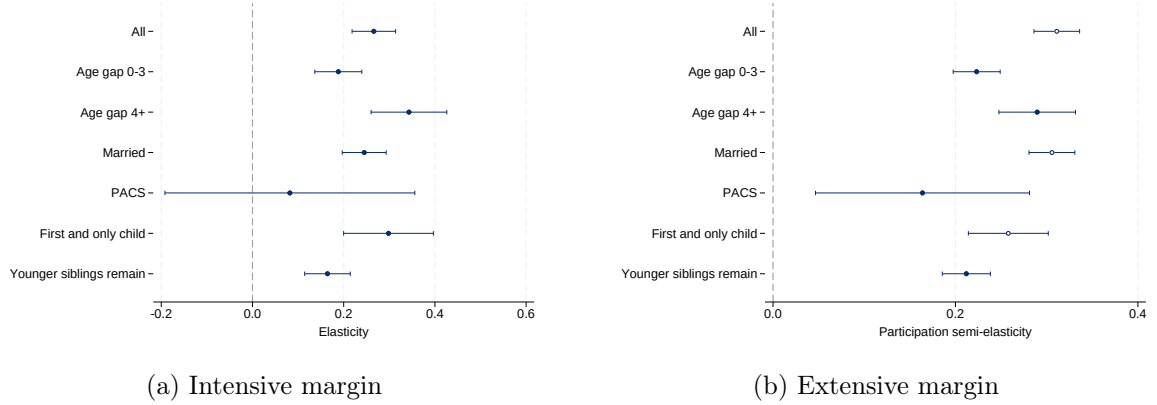


Figure 10: Elasticities by household subgroup

Notes: Pooled elasticities by subgroup: all cells of the subgroup are pooled as one aggregation group, with construction, screens and the hollow pre-trend flags following the display conventions of Appendix E.1; whiskers are 95% confidence intervals. Given the pooled samples' power, the pre-trend flags capture economically small differential trends. The “All” row reproduces the full estimation sample. Subgroups are defined at $r = -1$: the age gap in years between the parents; the legal union type; whether the exiting first-born is the only child ever observed in the tax unit. Estimates currently pool men only; the final version pools both genders (the women's subgroup estimates are pending). Appendix Figure E.8 reports the breakdown by gender.

5 On the relevance of accounting for heterogeneity

Section 4 documented substantial heterogeneity in estimated elasticities, both across the income distribution and by the composition of income within the household. In this section, we examine whether this heterogeneity matters for tax design. The analyses in this section are, at this stage, illustrative: they are run with placeholder elasticity inputs that mimic the shape of our findings, not yet with our estimates.

We assess this question against two methodologically distinct benchmarks. The first is the normative optimum: the Mirrlees–Saez tradition solves for the tax schedule that maximises a social welfare function subject to resource and incentive constraints (Mirrlees, 1971; Saez, 2001), and its counterpart for couples solves the corresponding multidimensional screening problem (Golosov & Krasikov, 2025; H. J. Kleven et al., 2009). The second is the evaluation of reforms: the perturbation approach of Bierbrauer et al. (2024) takes the observed tax system as given and derives sufficient statistics under which a move towards more individualised taxation of couples is Pareto- or welfare-improving. Section 5.1 conducts the first exercise

using our estimates of tax elasticities by own income; Section 5.2 conducts the second using our results on heterogeneity in tax responsiveness by the division of income within couples.

5.1 In optimal tax models

The optimality conditions of the Mirrlees–Saez framework are local: the optimal marginal rate at earnings level z is governed by the elasticity *at* z (Mirrlees, 1971; Saez, 2001). In the framework, a government chooses a tax schedule to maximise a social welfare function subject to its resource constraint and to incentive compatibility; because ability is unobserved, the schedule can condition on earnings but not directly on ability. Saez (2001) expresses the resulting optimality conditions in terms of empirically meaningful sufficient statistics: behavioural elasticities, the earnings distribution, and social marginal welfare weights. The optimal marginal rate at z balances the mechanical revenue and welfare effects of a small rate change against the behavioural response of taxpayers at z ; nothing in the formula requires the elasticity to be common across the distribution, and the constant-elasticity calibrations used in practice are a convenience rather than a feature of the theory.³²

We adapt the sufficient-statistics implementation of Brewer et al. (2010) and replace its common elasticity with a profile $e(z)$. Figure 11 provides an illustrative comparison. The dashed line assumes a constant elasticity of 0.326, the mean of the heterogeneous profile; the solid line instead uses a smooth elasticity profile that declines from 0.4 at low earnings to 0.1 at high earnings. Under both specifications the schedule is U-shaped over the displayed range. The comparison isolates the pointwise inverse-elasticity logic of the formula: the higher elasticity at low earnings lowers optimal marginal rates there, flattening the schedule at the bottom, while the lower elasticity higher in the distribution raises them—the schedule’s progressivity shifts up the income distribution. The flattening at low earnings is the part our estimates speak to directly; above the € 70,000 support of the estimates the elasticity profile is extrapolated, so the top of the schedule should not be read as a finding. A single elasticity calibrated to an economy-wide average accordingly misstates optimal rates at the bottom of the distribution in particular.

³²The sufficient-statistics formula of Saez (2001) already allows the elasticity to vary with earnings. We evaluate $e(z)$ pointwise while holding the earnings distribution, the social marginal welfare weights, and the revenue requirement fixed. The current figure uses an illustrative smooth profile declining from 0.4 to 0.1; in the final application, this profile will be replaced by an interpolation of our estimated income-group elasticities.

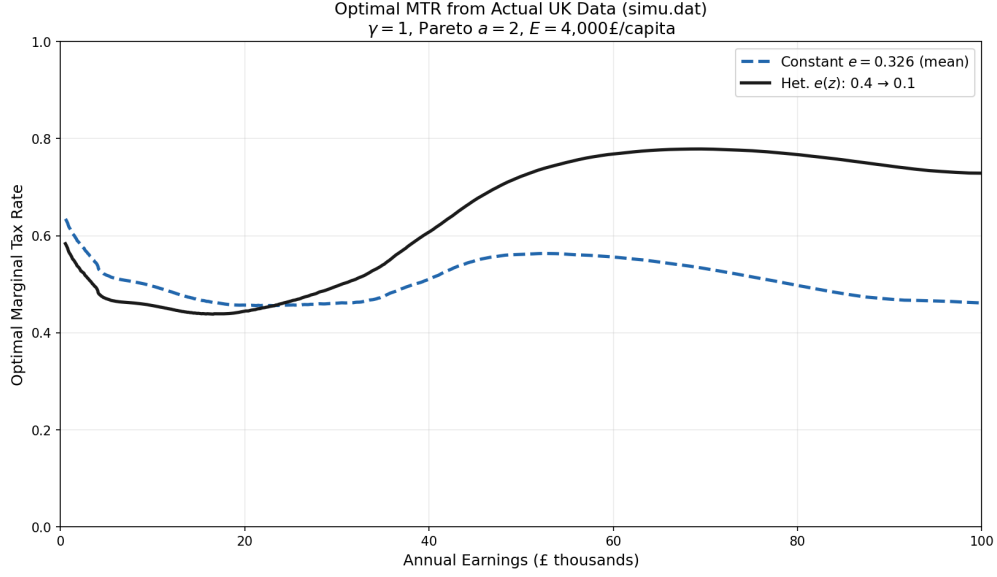


Figure 11: Optimal marginal tax rates under constant and heterogeneous elasticities

Notes: The figure plots optimal marginal tax rates from the sufficient-statistics implementation of Brewer et al. (2010). The dashed line assumes a constant elasticity of 0.326, the mean of the heterogeneous profile; the solid line uses an illustrative smooth profile declining from 0.4 to 0.1. The calibration uses the UK earnings distribution, logarithmic social welfare ($\gamma = 1$), a Pareto-tail parameter of $a = 2$, and a revenue requirement of £4,000 per capita.

This exercise exploits only the variation of responses with own earnings. By construction, a one-dimensional formula has no notion of a partner and therefore cannot absorb the variation of elasticities with the composition of household income. Using that dimension requires a framework for couples, to which we now turn.

5.2 In the assessment of jointness reforms

For couples, solving for the optimal two-dimensional tax schedule is a multidimensional screening problem, which is analytically demanding even under strong simplifying assumptions (Goloso & Krasikov, 2025; H. J. Kleven et al., 2009). In particular, the analytical characterisation in Goloso and Krasikov (2025) relies on a constant elasticity that is common to all individuals and therefore symmetric across spouses; its results would not survive elasticities that vary with the division of earnings within the couple, an extension that is, to our knowledge, work in progress. We therefore follow Bierbrauer et al. (2024), who evaluate reforms of the *existing* system rather than solving for the optimum. Starting from status-

quo couple taxation, they derive sufficient statistics that determine whether reforms towards more individualised taxation—reforms that allow the marginal rates of the two spouses to differ, lowering them on secondary relative to primary earnings in a revenue-neutral way—are Pareto-improving and, where they are not, whether they raise welfare under given welfare weights.

Their baseline calibration sets the elasticity of primary earnings to 0.25 and that of secondary earnings to 0.75. Our cell-specific estimates $\epsilon_{p,s}$ replace these two constants with an estimated surface: responses vary with the couple’s position in the joint distribution and with the division of earnings within it.

Figure 12 displays the welfare gains from moving towards a more individualised tax system under the constant-elasticity calibration (left column) and under our cell-specific estimates (right column). The first row assigns equal welfare weights to all households; the second uses near-Rawlsian weights that place greater weight on low-income households.

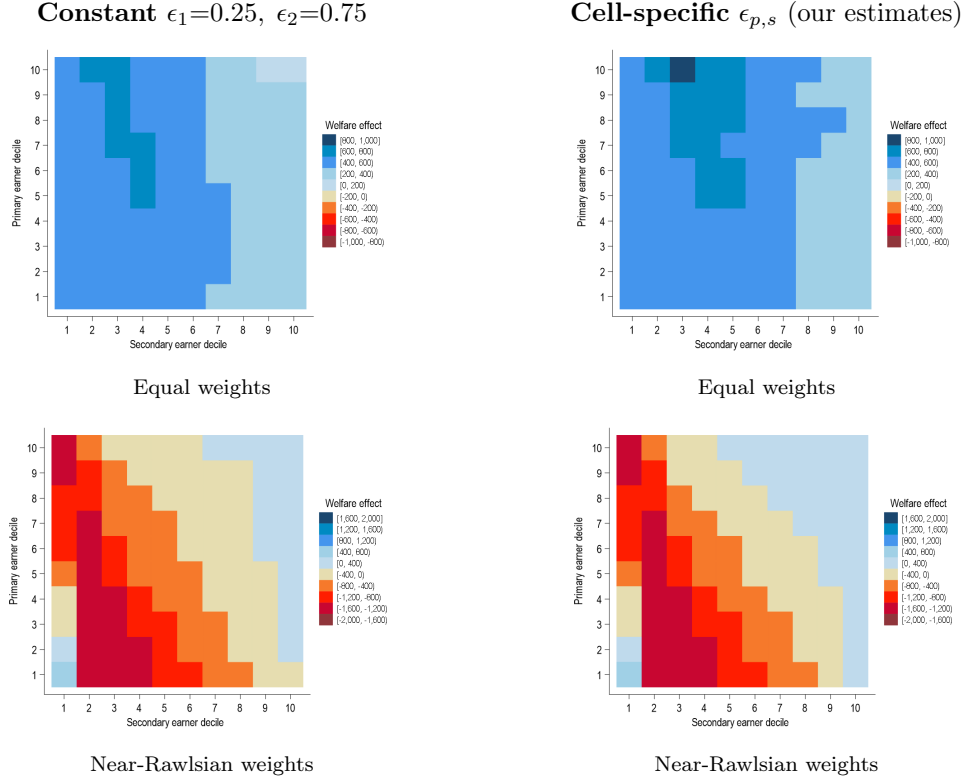


Figure 12: Welfare gains from a more individualised tax system

Notes: The left column uses constant elasticities of $\epsilon_1 = 0.25$ for primary earners and $\epsilon_2 = 0.75$ for secondary earners. The right column uses cell-specific elasticity estimates $\epsilon_{p,s}$. The top row assumes equal welfare weights, while the bottom row assumes near-Rawlsian welfare weights that place greater weight on low-income households. Comparing columns within each row shows that the assessment is sensitive to the interaction between the location of large elasticities in the joint distribution and the welfare weights: the cell-specific estimates raise the assessed gains from individualisation, most markedly under near-Rawlsian weights.

The two exercises show that accounting for heterogeneity changes the shape of the optimal tax schedule and the assessment of reforms towards more individual taxation of couples.³³³⁴³⁵

³³The direction of the optimal primary–secondary differential is itself unsettled. With a common elasticity for all individuals, optimal marginal rates on secondary earners are on average *higher* than on primary earners, because low secondary earnings identify couples to whom a redistributive planner wants to target transfers (Goloso & Krasikov, 2025); the classic argument of Boskin and Sheshinski (1983) instead prescribes *lower* rates on the more responsive spouse. Which force dominates depends on the size and location of the elasticity gap between spouses—the object we estimate.

³⁴Constant-elasticity calibrations are not an innocuous simplification in one dimension either: the welfare assessment of the post-war US tax cuts is sensitive to the assumed constant elasticity of taxable income (Bierbrauer et al., 2021); estimated elasticity profiles remove this ambiguity.

³⁵A further margin, which we do not pursue, is political feasibility: with heterogeneous elasticities, assessments of the political feasibility of moves towards individualisation may change (Bierbrauer et al., 2025).

6 Conclusion

This paper proposes a source of identifying variation for estimating tax elasticities, suited to mapping heterogeneities across gender, income and spousal income. Exploiting the institutional features of French family taxation, we use the fiscal exit of a couple’s first-born child—and, as a robustness design, marriage—as an event that changes the mapping from incomes to marginal tax rates, generating variation in marginal incentives through not-yet-tax-liable partner income, under transparently testable assumptions.

Using the universe of French tax declarations from 2003 to 2023 and a panel structure that follows individuals across household-unit changes, we show that headline averages—weighted to the French taxpayer distribution, 0.26 for men and 0.33 for women on the intensive margin, and 0.43 and 0.34 for participation—mask substantial heterogeneity. The averages are carried by low own incomes rather than by the remainder of the distribution, and, for women, where in the household the responses concentrate depends on the margin: their intensive-margin responses are largest where both partners earn little, their participation responses where the partner earns considerably more.

Replacing constant elasticities with profiles of the estimated shape changes the conclusions that standard frameworks for tax design reach: in the illustrative calibrations of Section 5, feeding the income profile into the sufficient-statistics formula shifts the schedule’s progressivity up the income distribution and flattens it at low incomes, relative to the constant-elasticity benchmark; and replacing the constant per-spouse elasticities in the evaluation of reforms towards more individual taxation of couples changes the predicted employment effects and raises the assessed welfare gains.

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A The household model and the interpretation of the estimand

This appendix relaxes the separability assumption of Section 3.1: it derives the household estimating equation under a general convex joint cost of earning, shows that joint filing collapses it back to equation (2) with the coefficient read as the elasticity of own income with respect to the household rate, and makes precise which elasticities the two events identify—including for the marriage design, which switches the filing regime.

We model a household choosing both spouses’ incomes against a joint budget constraint,

$$\max_{z_{it}, z_{pt}} c_t - \Psi(z_{it}, z_{pt}; n_{it}, n_{pt}) \quad \text{s.t.} \quad c_t \leq z_{it} + z_{pt} - \mathcal{T}_t(z_{it}, z_{pt}),$$

where z_{pt} is the partner’s income, $\mathcal{T}_t$ is the couple’s tax function—whose form depends on the filing regime (Section 2.1), a dependence our design uses and reform settings usually abstract from—and Ψ is a convex joint cost of earning income. The joint cost may be additively separable across spouses or non-separable, reflecting leisure complementarities, home production, or coordinated retirement (Gelber, 2014; Goux et al., 2014).³⁶

The potential incomes n_{it} and n_{pt} index each spouse’s cost of earning income; we normalise them so that n_{it} is the income i would earn if both spouses faced marginal tax rates of zero—as in the separable case of Section 3.1.

Writing $\tau_{it} \equiv 1 - \partial \mathcal{T}_t / \partial z_{it}$ and $\tau_{pt} \equiv 1 - \partial \mathcal{T}_t / \partial z_{pt}$ for the spouses’ marginal net-of-tax rates, the first-order conditions are $\partial \Psi / \partial z_{it} = \tau_{it}$ and $\partial \Psi / \partial z_{pt} = \tau_{pt}$. Totally differentiating the pair around the pre-event optimum and solving the resulting two-equation system yields the elasticities

$$(A.1) \quad \epsilon_{ii} = \frac{\tau_{it}}{z_{it}} \frac{\Psi_{pp}}{D}, \quad \epsilon_{ip} = -\frac{\tau_{pt}}{z_{it}} \frac{\Psi_{ip}}{D}, \quad D \equiv \Psi_{ii} \Psi_{pp} - \Psi_{ip}^2 > 0,$$

³⁶Quasi-linearity in household consumption shuts down income effects at the household level: neither the partner’s net income nor the change in the household’s tax *liability* affects income choices, and it makes the intra-household allocation of consumption irrelevant for those choices: with transferable utility, any sharing rule supported by lump-sum transfers leaves the first-order conditions unchanged. For the same reason the model’s predictions for incomes coincide with those of an efficient collective household à la Chiappori (1992) with quasi-linear private utilities—we need not take a stance on the household decision process. Gelber (2014) rejects the income-pooling restriction of the unitary model; the rejection operates through income effects, which quasi-linearity shuts down, so the income predictions used here do not rest on the rejected restriction. See Appendix C for the extension with income effects.

where subscripts on Ψ denote second derivatives with respect to the spouses' incomes, evaluated at the optimum, and $D > 0$ by strict convexity—and the household estimating equation in changes,³⁷

$$(A.2) \quad \Delta \ln z_{it} = \epsilon_{ii} \Delta \ln \tau_{it} + \epsilon_{ip} \Delta \ln \tau_{pt} + \Delta \ln n_{it}.$$

Every coefficient in (A.2) is an elasticity of i 's *own* income; the subscripts index whose rate varies: ϵ_{ii} is the response of i 's income to i 's own rate, ϵ_{ip} its response to the *partner's* rate. Both are derivatives of the household's optimal income-supply function $z_i^*(\tau_{it}, \tau_{pt})$ —jointly re-optimised responses, in which the partner's income also adjusts—and the cross-derivative Ψ_{ip} governs both: it is the numerator of ϵ_{ip} , and it enters ϵ_{ii} through D , so that even when i 's rate alone changes, the partner's re-optimisation raises ϵ_{ii} above the one-person elasticity $\tau_{it}/(z_{it}\Psi_{ii})$, by the factor $1/(1 - \rho^2)$ with $\rho \equiv \Psi_{ip}/\sqrt{\Psi_{ii}\Psi_{pp}}$. The split between ϵ_{ii} and ϵ_{ip} is thus by price, not by mechanism. Under quasi-linearity the cross-rate response is transmitted entirely through the partner's behavioural adjustment, and it is zero when effort costs are separable, $\Psi_{ip} = 0$.³⁸

Under joint filing, the marginal rate is common to both spouses, $\tau_{it} = \tau_{pt} = \tau_{it}^H$ (Section 2.1), so any change in it is common as well, and equation (A.2) collapses to the estimating equation (2) of the main text,

$$\Delta \ln z_{it} = \underbrace{(\epsilon_{ii} + \epsilon_{ip})}_{\equiv \epsilon_i^H} \Delta \ln \tau_{it}^H + \Delta \ln n_{it},$$

with the coefficient read as the *elasticity of own income with respect to the household rate*, $\epsilon_i^H \equiv \epsilon_{ii} + \epsilon_{ip}$: the response of own income to the common rate, including the component transmitted through the partner's adjustment. Explicitly, from (A.1), $\epsilon_i^H = (\tau_{it}^H/z_{it})(\Psi_{pp} - \Psi_{ip})/D$: it nests the one-person elasticity at $\Psi_{ip} = 0$ and exceeds it under complementarity ($\Psi_{ip} < 0$) through both channels, the cross term and the feedback.

³⁷For additively separable, iso-elastic Ψ the relation is exact and $\epsilon_{ip} = 0$; in general, the elasticities are local to the pre-event optimum. The error term is composite: with non-separable costs, changes in the *partner's* potential income also shift i 's income—they enter through the same cross-derivative Ψ_{ip} —and $\Delta \ln n_{it}$ stands for the combined contribution of both spouses' potential-income changes to i 's income. That this component does not vary with the dose within comparison cells is what Assumption 1 restricts; the model does not deliver it.

³⁸Separability is the assumption underlying most quantitative work on couples taxation (Bick & Fuchs-Schündeln, 2018).

Under non-separable costs, all three elasticities are couple-level parameters: both spouses' cost curvatures enter each of them. They are local to the couple—the same individual would exhibit a different elasticity if single—and estimates from couples are not directly comparable to estimates on samples of singles. A gradient of responses in partner income at fixed own income can accordingly arise from three sources: sorting of individual elasticities across partner types, income effects (Appendix C), or the non-separability studied here; the design identifies the gradient, not its decomposition.

At the child's fiscal exit, the household files jointly before and after the event, so the event-induced rate change is common to both spouses, and the design recovers the elasticity of own income with respect to the household rate, ϵ_i^H .³⁹

At marriage, by contrast, spouses are individually taxed before the event, so the two spouses' predicted changes generally differ—often only one spouse is predicted to see any change. Among couples for whom the event predicts a change in i 's rate only, the event-induced variation moves $\Delta \ln \tau_{it}$ alone, and the reduced form identifies the own-rate elasticity ϵ_{ii} —itself inclusive of the partner's re-optimisation (the factor $1/(1 - \rho^2)$ above). Among the same couples, the *partner's* income response to i 's rate change identifies the cross-rate elasticity ϵ_{pi} directly—the design of Gelber (2014) reproduced within our setting. The current implementation does not condition on the partner's predicted change: among treated individuals the partner—typically the higher earner—may simultaneously be predicted to cross a threshold downwards, in which case the reduced form mixes ϵ_{ii} with a term in ϵ_{ip} . The contamination is bounded by the cross-rate elasticity—small by the estimates cited below; imposing the restriction is a candidate refinement for a rerun of the marriage design.

The gap between ϵ_i^H and ϵ_{ii} is likely small: exploiting the Swedish tax reform of 1990–1991, in which spouses' marginal rates moved independently, Gelber (2014) estimates compensated cross-rate elasticities of 0.05 for husbands and 0.07 for wives, an order of magnitude below his own-rate estimates of 0.41 and 0.47—and positive, so that ϵ_i^H weakly overstates ϵ_{ii} . Where

³⁹The income-weighted average of ϵ_i^H across spouses gives a different object: the response of the couple's *combined* income to the household rate, the one relevant for revenue and Laffer calculations—whichever way the response splits across own- and cross-rate channels.

the distinction matters, the marriage design provides internal estimates, and external ones can be imputed instead: $\epsilon_{ii} = \epsilon_i^H - \hat{\epsilon}_{ip}$.⁴⁰

B Proof of Proposition 1

Proof. By (4), (5) and Assumption 2, $\Delta y_{ir} = \Delta_r Y_i(0) + \mathbb{1}[Z_{ir} \neq 0] \epsilon_i \Phi_{ir}$ and $\Delta \text{NTR}_{ir} = \Delta_r \text{NTR}_i(0) + \mathbb{1}[Z_{ir} \neq 0] \Phi_{ir}$. By Assumption 1, the conditional means of the untreated components are equal for the $Z_i = \delta$ and $Z_i = 0$ groups of the same cell, so they cancel in the contrasts: $RF_c^r(\delta) = \mathbb{E}_c[\epsilon_i \Phi_{ir} \mid Z_i = \delta]$ and $FS_c^r(\delta) = \mathbb{E}_c[\Phi_{ir} \mid Z_i = \delta]$. Taking the ratio gives the weighted-average representation, with weights averaging to one by construction; under Assumption 3 all Φ_{ir} are non-positive, hence share the sign of their conditional mean, and the weights are non-negative. $\square$

⁴⁰In more detail, two routes lead from the identified total to the structural own-rate elasticity. **Route 1** (*spousal separability*): with $\Psi_{ip} = 0$, $\epsilon_{ip} = 0$ and $\epsilon_i^H = \epsilon_{ii}$; specialising to the iso-elastic form, the household problem separates into two copies of the one-person model, each facing the household rate—the formal justification for the standard practice of pairing the one-person model with household-rate variation. **Route 2** (*external adjustment*): impute $\hat{\epsilon}_{ip}$ from Gelber (2014) and report $\epsilon_{ii} = \epsilon_i^H - \hat{\epsilon}_{ip}$ as an adjustment or as bounds; relative to our estimates, which decline with income, the implied adjustment is modest but not uniformly negligible, and we report adjusted series where the distinction matters. The imputation matches the model’s object: because the model is quasi-linear, ϵ_{ip} is a compensated cross-rate elasticity, and Gelber’s estimates are compensated as well. The uncompensated added-worker channel operates through income effects—absent here, treated in Appendix C—and is opposite-signed and potentially large after major income shocks (Blundell et al., 2016); for bracket-switch variation, the associated liability changes are small relative to lifetime resources, which confines it to second order. Gelber’s estimates remain gender-specific and from a different population and margin, so we treat them as orders of magnitude rather than point corrections, and apply the adjustment to the intensive margin only. The distinction matters most for the heterogeneity results: gender differences in ϵ_i^H are preference parameters only under Route 1, or after the Route 2 adjustment; without either, part of a measured gender gap can reflect the other spouse’s adjustment transmitted within the household—mechanically so, since symmetry of Ψ restricts the cross responses in levels, not in elasticities: $\epsilon_{ip}/\epsilon_{pi} = \tau_{pt}z_{pt}/(\tau_{it}z_{it})$, the income ratio under a common rate, so the lower earner’s cross-rate elasticity is the larger one (consistent with Gelber’s larger estimate for wives), and the adjustment removes exactly this component. Because we estimate both spouses’ responses to the same household shock within the same cells, the pair of estimates is itself informative about such reallocation, and we return to this when interpreting the heterogeneity results.

C Model extensions

C.1 Income effects

Relaxing quasi-linearity, we replace linear consumption with a concave utility $v(c_t)$: the household problem becomes

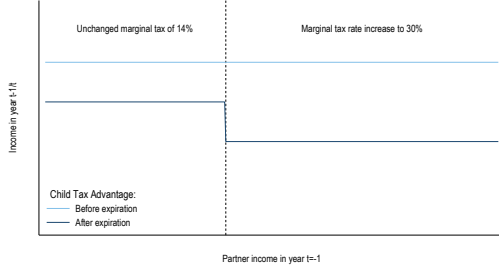
$$\max_{z_{it}, z_{pt}} v(c_t) - \Psi(z_{it}, z_{pt}; n_{it}, n_{pt}) \quad \text{s.t.} \quad c_t \leq z_{it} + z_{pt} - T_t^H(z_{it} + z_{pt}; Q_{it}) - \kappa_{it},$$

where κ_{it} collects lump-sum household expenses, which may change at the event (expenses for the child). The first-order conditions become

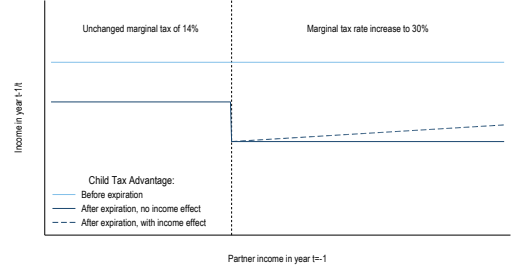
$$\frac{\partial \Psi}{\partial z_{jt}} = v'(c_t) \tau_{jt}^H, \quad j \in \{i, p\} :$$

income choices now depend on the marginal utility of household consumption, and therefore on partner income, on the level of the tax *liability*, and on needs κ_{it} . Log-linearising adds a virtual-income term to equation (2), in the spirit of Gruber and Saez (2002). Two remarks follow for our design. First, households predicted to cross a bracket at the event also experience a larger increase in tax liability than their within-cell controls, so, absent a virtual-income instrument, our estimates should be read as total responses to the event-induced tax change. The two channels oppose each other—the higher marginal rate reduces income, while the larger liability raises it when leisure is normal—so under income effects our estimates are lower bounds on the compensated elasticity; an extension instrumenting a simulated virtual-income term would separate the two responses. Second, the needs change is part of the event, not of the schedule change: the child leaves, and the expenses change, under both schedules (Section 3.3), so the income consequence of the needs change belongs to untreated growth and is restricted by Assumption 1 rather than Assumption 2. The classification has teeth: with concave v , even a needs change common to every household of a cell moves income through $v'(c_t)$, and consumption varies with partner income within the cell’s band—income effects can thus convert a common event shock into a partner-income *gradient* in growth. This is exactly the class of violation the validation region detects.⁴¹

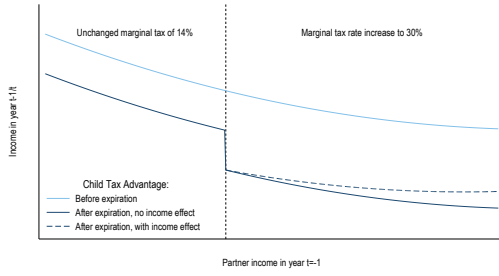
⁴¹Appendix G takes income effects seriously for a different purpose: they are one reason why first-stage monotonicity can fail in reform-based simulated-IV designs, though not in ours.



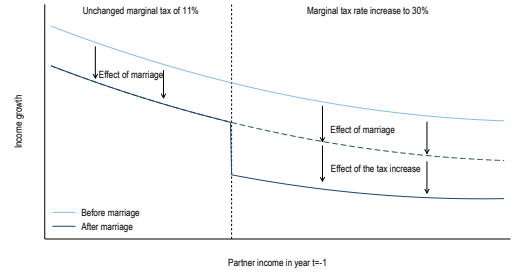
(a) Child's fiscal exit, linear schedule, no income effects



(b) Child's fiscal exit, linear schedule, income effects



(c) Child's fiscal exit, progressive schedule



(d) Marriage, with income growth

Figure C.1: Income effects

Notes: Budget-set illustrations of the event-induced schedule change and the separation of income and substitution effects.

C.2 The participation margin

The intensive-margin model cannot generate extensive-margin responses. Returning to the quasi-linear case, and following H. J. Kleven et al. (2009), consider a spouse deciding whether to work at all, facing a fixed cost of participation q_{it} (commuting, childcare, forgone home production). Conditional on working, income is chosen optimally, so the value of participation is

$$V_{it}^w = \max_z \left\{ z - [T_t^H(z + z_{pt}; Q_{it}) - T_t^H(z_{pt}; Q_{it})] - \psi(z; n_{it}) \right\},$$

where the bracketed term is the *participation tax* (Immervoll et al., 2007): the increase in the household's tax liability caused by the spouse's income, given partner income and the household's tax schedule. The spouse works if and only if

$$V_{it}^w \geq q_{it}.$$

The relevant price on this margin is therefore not the marginal rate at current income but the participation tax wedge—and its first-euro marginal rate, $1 - \tau^H$ evaluated at $z = 0$, is exactly what partner income and the quotient determine: because partner income alone places the household above the tax-free threshold, the second earner’s first euro is taxed at the household’s current bracket. A household pushed into a higher bracket by the child’s fiscal exit thus sees the return to participation fall, and with heterogeneous fixed costs q_{it} the participation probability falls with the participation tax—the sign of the extensive-margin responses we document. The estimates themselves, however, are not yet expressed in this price: the implementation instruments the same object on both margins, the event-induced change in the household *marginal* net-of-tax rate evaluated at current own income, whereas the wedge above is governed by the schedule between z_{pt} and $z_{it} + z_{pt}$, its first-euro rate by partner income and the quotient alone. Within cells the two move closely together—the predicted bracket shift raises the inframarginal burden over most of the income range—so we read the extensive-margin estimates as participation responses to the marginal-rate change, rescaled by pre-event participation (Section 3.4); converting them into elasticities with respect to one minus the participation tax rate is a pending design decision.

D Complementary material: case study

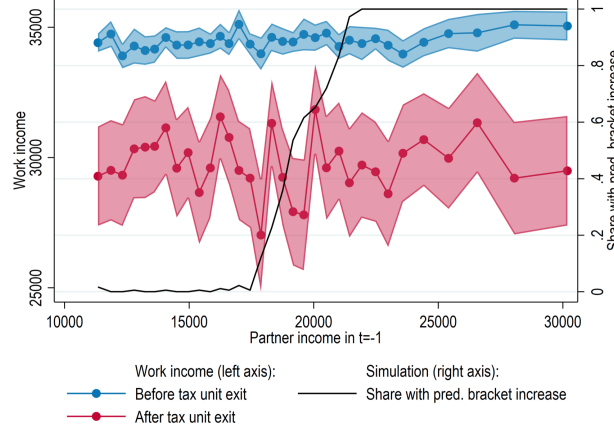


Figure D.1: Identifying variation at the first-born's fiscal exit: men

Notes: Male counterpart of Figure 5. Sample: men in married or civil-union tax units whose first-born exits the tax unit between 2003 and 2023. The probability of a predicted tax increase rises sharply with partner income, dividing the sample into a validation region and a treatment region; see the notes to Figure 5 for construction details.

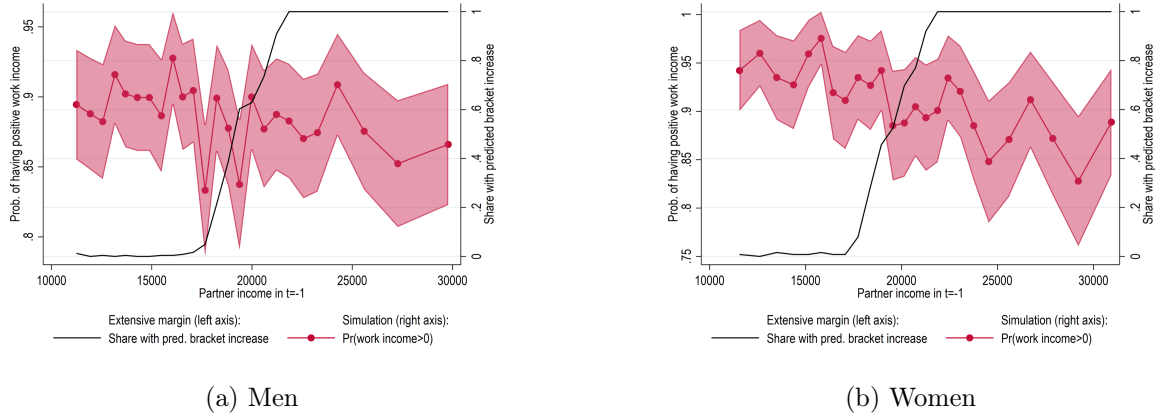
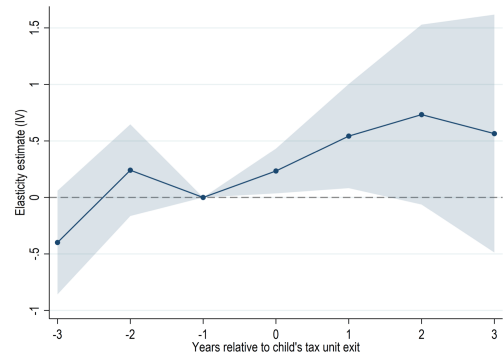


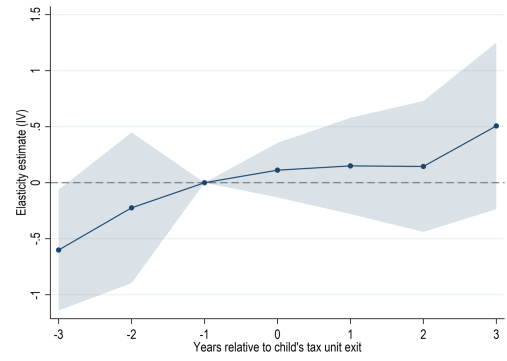
Figure D.2: Identifying variation at the first-born's fiscal exit: extensive margin

Notes: Extensive-margin counterpart of Figures 5 and D.1: the outcome is an indicator for positive work income. Sample and construction as in Figure 5.

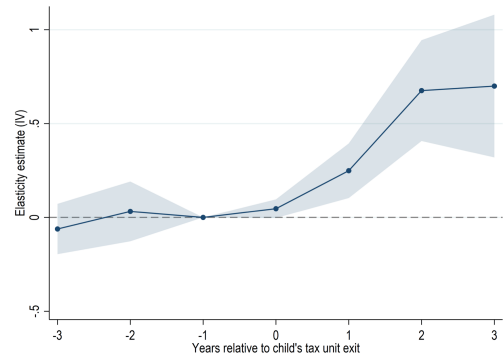
Translating these earnings paths into elasticities with the estimation approach of Section 3 yields the second-stage estimates shown in Figure D.3.



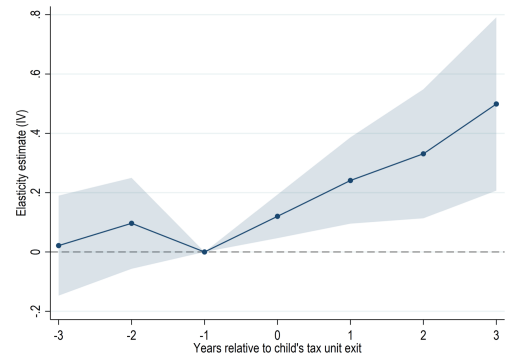
(a) Intensive, women



(b) Intensive, men



(c) Extensive, women



(d) Extensive, men

Figure D.3: Second-stage estimates for the case-study cell

Notes: Two-stage least squares estimates of the elasticity of own income with respect to the household net-of-tax rate for the case-study example of Figure 4, by event year r relative to the first-born's fiscal exit, with $r = -1$ the omitted reference; shaded areas are 95% confidence intervals. The upper panels report the intensive margin and the lower panels the extensive margin, men on the left and women on the right. Sample: the example cell (pre-event own income in $[\text{€}34,000, \text{€}36,000]$, pre-event marginal rate 14%, 2.5 fiscal parts, first-born exiting at $r = 0$). The instrument, screens and construction follow Section 3.

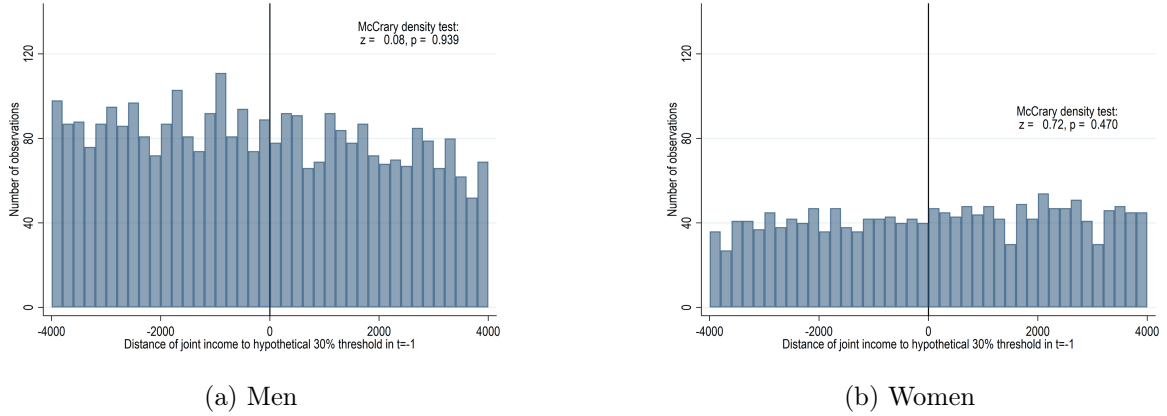


Figure D.4: McCrary density test at the hypothetical joint-income threshold

Notes: Each panel plots the number of observations by the distance of pre-event joint income to the hypothetical threshold at which the marginal tax rate rises from 14% to 30% once the first-born’s exit reduces the family quotient, for the case-study example of Figure 4, separately for men in Panel (a) and women in Panel (b). This distance is the running variable underlying the binary instrument: households above the threshold are predicted treated, households below it predicted control. If households anticipated the exit and adjusted incomes strategically—bunching below the future threshold, or timing the detachment—the pre-event density would show excess mass around zero. The distribution is flat through the threshold and the McCrary test does not reject continuity ($z = 0.08$, $p = 0.94$ for men; $z = 0.72$, $p = 0.47$ for women).

E Complementary material: results section

E.1 Display conventions for the cell-level exhibits

The headline exhibits of Section 4 share a two-layer display convention, applied to the two-stage least squares cell–dose estimates of Section 3.5 (equation (6)) on the status-stable sample, pooled over event years $r = 0$ to 3 with the frequency weights of Section 3.4. The standard error of a displayed aggregate is the correspondingly weighted mean of the cell-level delta-method standard errors, treating cells as independent.

Cell screens. A cell–dose–year estimate enters a displayed aggregate if its elasticity is at most 5 in absolute value, its outcome-scale standard error lies in $(0, 1]$, its first stage has $|t| \geq 3$ (where a first stage is exported), and its elasticity-scale standard error is at most 0.5 for the income profiles, event-time and subgroup exhibits, and at most 0.4 for the maps—tighter because a tile prints a colour without a displayed confidence interval.

Aggregate-level rules. The maps additionally display a tile only if its pooled standard error is at most 0.4 and at least 1,000 treated individuals underlie it, counting each individual once

across event years, and show the income range supported for both genders (up to €60,000). Profiles and subgroup exhibits apply no aggregate-level screens: imprecise aggregates appear with wide confidence intervals rather than being suppressed.

Pre-trend annotation. For every displayed aggregate, the pre-event placebo reduced form (Section 3.5) is pooled over $r = -3$ and -2 ; aggregates with $|t| \geq 1.96$ are crossed out (maps) or drawn hollow (profiles and subgroup exhibits). The flag is an annotation: no estimate is dropped or altered on account of pre-trends.

E.2 Balance across predicted groups

The balance of predetermined characteristics behind the pooled validation of Section 4 is summarised in Table E.1: means for the predicted-control and predicted-treated groups and the standardised mean difference averaged across comparisons. The two groups are closely comparable on the outcome’s own pre-event dynamics—income growth and participation differ by less than 0.05 of a standard deviation. Differences of 0.2 to 0.3 appear in own and partner income levels, in age, and in pension receipt: within an own-income bin, the predicted treated are those whose household income sits higher in the bin, and these are the correlates of household taxable income, which determines assignment. The largest differences are in household taxable income itself and in family size, which shifts the number of fiscal parts and hence the predicted rate change.

Table E.1: Balance across predicted-treated and predicted-control comparisons

	Women			Men		
	Control	Treated	SMD	Control	Treated	SMD
<i>Pre-event outcomes</i>						
Work income, $r = -1$ (€)	16,558	17,167	0.23	27,116	27,710	0.23
Log work-income growth, $r = -3$ rel. $r = -1$ (%)	-2.35	0.68	0.04	11.24	14.28	0.03
Positive work income, $r = -1$ (%)	79.4	81.0	0.03	89.4	89.1	-0.01
<i>Composition</i>						
Partner work income, $r = -1$ (€)	25,783	28,382	0.24	16,085	17,679	0.30
Partner has positive work income, $r = -1$ (%)	84.6	87.2	0.04	77.3	82.3	0.10
Receives pension income, $r = -1$ (%)	3.9	6.7	0.12	7.7	10.9	0.13
Age	49.5	50.6	0.25	51.8	52.8	0.22
Partner age	52.1	52.9	0.17	49.4	50.6	0.28
Number of dependent children	2.24	1.97	-0.47	2.29	1.95	-0.56
Fiscal parts, $r = -1$	3.37	3.15	-0.46	3.41	3.13	-0.55
<i>Assignment and first stage (differ by design)</i>						
Household taxable income, $r = -1$ (€)	39,584	45,112	1.06	39,596	44,416	1.04
Predicted change in log net-of-tax rate ($\times 100$)	0.00	-11.42	-	0.00	-11.30	-
Realised net-of-tax-rate change, $r = +1$ (pp)	-1.36	-5.56	-	-1.49	-5.59	-
<i>Sample</i>						
Unique individuals	842,950	534,851		836,555	512,991	
Comparison-year cells		964			1,215	

Notes: Each treated tax-change dose is compared with the zero-change controls in the same own-income bin, partner-income tercile, baseline bracket, and event year. Means are computed within each comparison-year arm and averaged with equal weight across comparison-year cells; SMD is the equally weighted mean of the signed within-cell standardised differences. Household taxable income determines assignment and differs between arms by construction (standardised difference 1.06 for women, 1.04 for men). The predicted and realised rate changes are zero in the control arm by construction, so no standardised difference is reported for them. The log-income-growth row requires positive work income at $r = -1$ and at $r = -3$. Controls may enter several dose comparisons; the counts report unique individuals. Levels are in euros of the event year, not deflated. Sample as in Figure 6. The complete table—all balance variables, the 90th percentile of absolute standardised differences, and per-row comparison counts—is available on request.

E.3 Density tests at the threshold, pooled

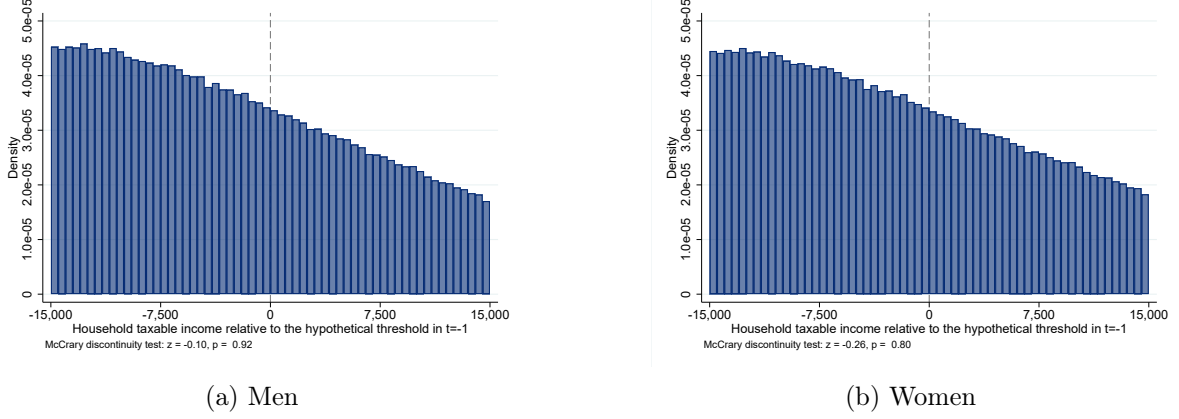


Figure E.1: Density tests at the hypothetical threshold, pooled across cells

Notes: Density of household taxable income at $r = -1$, measured relative to each comparison cell's hypothetical post-event threshold and pooled across the comparison cells of the design; unlike the observation counts of Figure D.4, the vertical axis is a density. The estimated log-density discontinuities are -0.002 (s.e. 0.010, $p = 0.80$, $N = 782,811$) for women and -0.001 (s.e. 0.009, $p = 0.92$, $N = 776,649$) for men: no evidence of sorting of the event's timing at the threshold.

E.4 Graphical evidence: event-time profiles

Event-time profiles of the estimates will be reported here (exhibits temporarily removed pending a data check).

E.5 Aggregated elasticities

Weighted to the population of French taxpayers, the average participation response is 0.386 (0.342 for women, 0.430 for men); across the four weighting schemes the average lies between 0.20 and 0.39. The gender ordering reverses relative to the intensive margin: men's participation responses exceed women's under every scheme, whereas women's intensive-margin elasticities exceed men's.

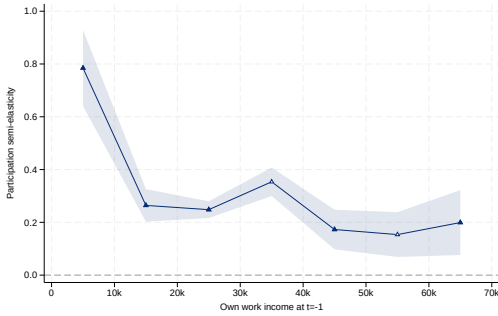
Table E.2: Aggregated extensive-margin elasticities

	Men	Women	Total
Population-weighted: French taxpayers	0.430 (0.026)	0.342 (0.023)	0.386 (0.017)
Population-weighted: event population	0.392 (0.021)	0.267 (0.026)	0.329 (0.017)
Inverse-variance weighted	0.220 (0.010)	0.172 (0.010)	0.196 (0.007)
Income-weighted	0.275 (0.013)	0.262 (0.016)	0.269 (0.010)

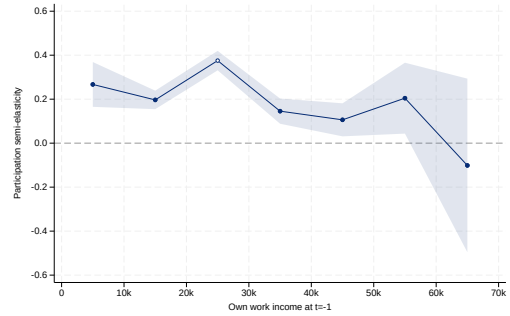
Notes: As Table 5, for the extensive margin: the outcome is an indicator for positive work income, and the entries are elasticities of participation with respect to the household net-of-tax rate—the cell coefficients rescaled by pre-event participation (see the footnote in Section 3.4).

E.6 Maps by own income

The extensive-margin counterparts of the own-income displays of Section 4.2.2. The outcome is an indicator for positive work income; bin estimates are participation (semi-)elasticities constructed as described in Section 3. Samples, pooling window ($r = 0$ to 3) and screens are identical to the intensive-margin displays throughout; axis ranges adapt to each panel.



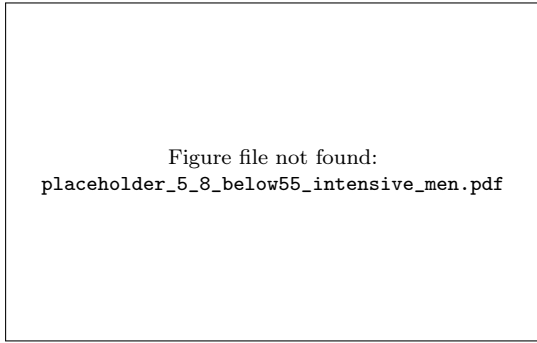
(a) Men



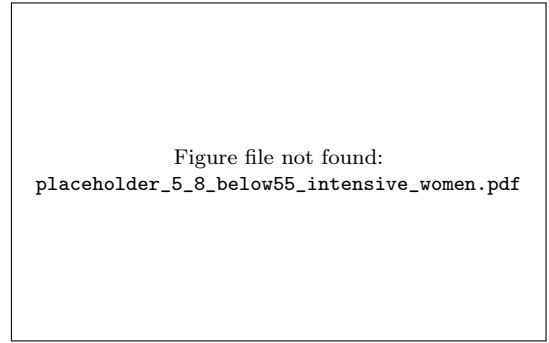
(b) Women

Figure E.2: Extensive-margin elasticities along the own-income distribution

Notes: As Figure 8, for the extensive margin: the outcome is an indicator for positive work income and the estimates are participation responses. Two differences from the intensive-margin profile are worth noting: the men's €0–10,000 bin, which is not estimable on the intensive margin, carries their largest participation response, and the women's profile extends one bin further, to €60,000–70,000.



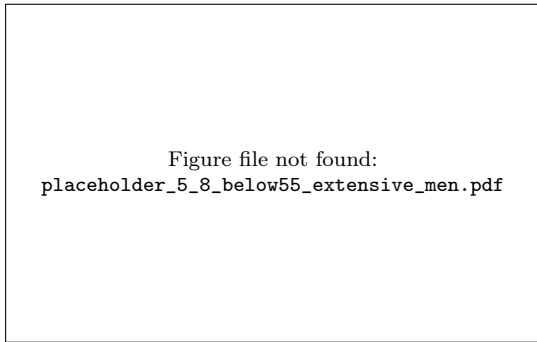
(a) Men



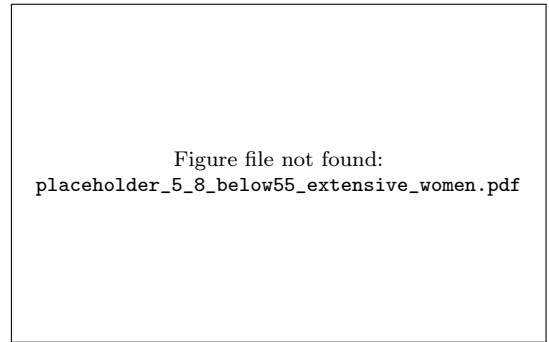
(b) Women

Figure E.3: Intensive-margin profiles: all ages versus below 55

Notes: Exhibit on hold: the panels are placeholders pending a data check. The completed run overlays the headline profile (all ages, dark navy, solid) with the profile estimated on individuals below age 55 at the event (light blue, dashed); the values discussed in Section 4.2.2 are women's lowest-bin elasticity falling from 0.62 (all ages) to 0.38 (below 55) with the declining shape intact, and the men's below-55 intensive profile indistinguishable from zero in every bin. Otherwise as Figure 8. Samples, screens, pooling window and construction are identical to Figure 8; shaded areas are 95% confidence intervals in the colour of the corresponding series, and hollow markers flag the bin-level pre-trend test of the corresponding series. The extensive-margin counterpart is Figure E.4.



(a) Men



(b) Women

Figure E.4: Extensive-margin profiles: all ages versus below 55

Notes: As Figure E.3, for the extensive margin, and on hold with it. In the completed run the lowest-earning men below 55 retain a large but imprecise participation response (1.40, s.e. 0.32), while participation responses above the bottom bin fade for both genders.

E.7 Maps by own and partner income

Tables E.3 and E.4 report the estimate, standard error and number of underlying cell-dose-period estimates for every tile of the heterogeneity maps, both margins.

Table E.3: Tile-level estimates: women

Own income	Partner tercile	Intensive		Extensive	
		Estimate	Cells	Estimate	Cells
0–10k	low	0.71 (0.17)	4	0.14 (0.08)	20
	mid	0.44 (0.27)	1	0.42 (0.07)	15
	high	0.17 (0.17)	3	0.52 (0.09)	6
10–20k	low	0.01 (0.05)	23	-0.08 (0.03)	38
	mid	0.14 (0.09) [†]	11	0.85 (0.07)	30
	high	0.09 (0.06)	10	0.17 (0.02)	26
20–30k	low	0.07 (0.04) [†]	22	0.46 (0.05)	39
	mid	0.40 (0.07)	11	0.45 (0.04) [†]	30
	high	0.16 (0.05)	12	0.13 (0.02)	22
30–40k	low	0.13 (0.08)	10	0.20 (0.05)	28
	mid	0.03 (0.07)	10	0.13 (0.03)	14
	high	0.15 (0.11)	5	0.08 (0.06)	9
40–50k	low	0.35 (0.10) [†]	6	0.10 (0.04) [†]	11
	mid	0.29 (0.18)	2	0.07 (0.09)	8
50–60k	low	0.06 (0.19)	1	0.21 (0.08)	7

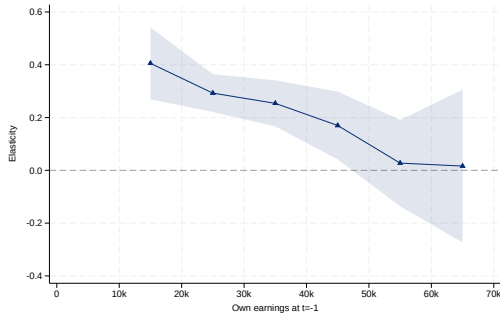
Notes: One row per own-income bin (thousand euro) $\times$ within-bin partner-income tercile of Figure 9, both margins. Estimates follow the map-level display conventions of Appendix E.1 and match the printed tiles exactly; standard errors in parentheses are the correspondingly weighted means of the cell-level delta-method standard errors, treat cell estimates as independent, and are mildly understated, as cells within a rectangle share control observations across periods. Daggers mark rectangles failing their pooled pre-period reduced-form placebo ($|t| \geq 1.96$), matching the crossed-out tiles of the maps; dashes mark tiles removed by the display screens on one margin, and rectangles displayed on neither margin are omitted. Cells columns give the number of underlying cell-dose-period estimates; with the pooling window $r = 0$ to 3, one cell-dose can contribute up to four.

Table E.4: Tile-level estimates: men

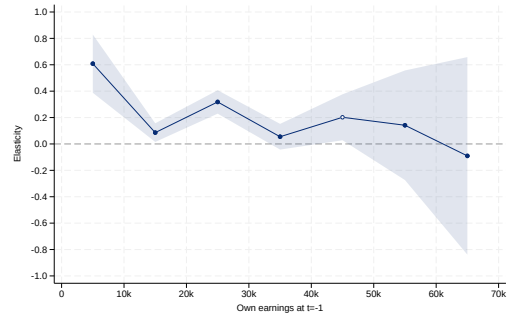
Own income	Partner tercile	Intensive		Extensive	
		Estimate	Cells	Estimate	Cells
0–10k	low	–	–	1.18 (0.20)	3
	mid	–	–	0.37 (0.12)	6
	high	–	–	0.56 (0.11)	4
10–20k	low	0.36 (0.12)	7	0.27 (0.05)	26
	mid	0.18 (0.09)	10	0.02 (0.04)	24
	high	0.37 (0.14) [†]	5	0.39 (0.06)	19
20–30k	low	0.26 (0.04)	28	0.13 (0.02)	43
	mid	0.78 (0.11)	16	0.42 (0.05)	30
	high	0.29 (0.07)	13	0.34 (0.03)	29
30–40k	low	0.38 (0.08)	17	0.19 (0.04)	34
	mid	0.26 (0.08)	9	0.51 (0.04)	19
	high	0.21 (0.06) [†]	11	0.14 (0.02) [†]	19
40–50k	low	0.01 (0.13)	6	0.28 (0.06) [†]	15
	mid	0.10 (0.08)	6	0.17 (0.04)	14
	high	0.00 (0.12)	5	0.01 (0.06)	11
50–60k	low	0.13 (0.11)	6	0.22 (0.06)	14
	mid	0.19 (0.11)	3	0.01 (0.07) [†]	9
	high	0.22 (0.28)	1	0.07 (0.08)	5

Notes: As Table E.3, for men. The €20,000–30,000 mid-tercile intensive tile (0.78) is dominated by a single comparison whose dose group fails a disaggregated pre-period test; the rectangle-level placebo does not flag it, so it carries no dagger (Section 4.2.3).

E.8 Alternative outcomes



(a) Men



(b) Women

Figure E.5: Intensive-margin elasticities with earnings as outcome

Notes: As Figure 8, with earnings (work plus self-employment income) as the outcome and own-income bins on earnings at $r = -1$, from the status-stable earnings estimation; construction, screens and the hollow pre-trend flags follow the display conventions of Appendix E.1. The earnings profiles closely track the work-income profiles. Intensive margin only: no extensive-margin earnings estimate survives the cell screens.

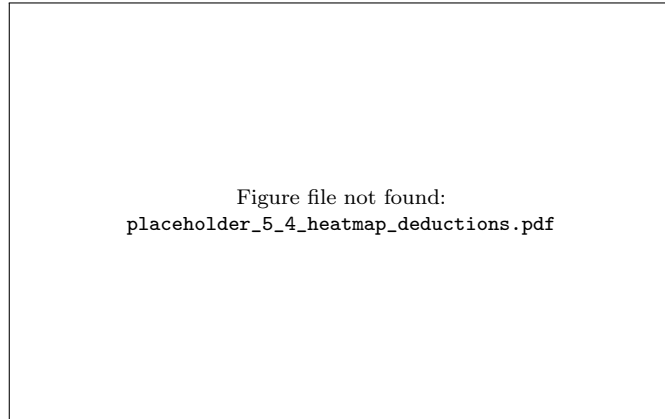


Figure E.6: Responses by own and partner income: deductions as outcome

Notes: Placeholder: no estimates are shown. The figure will report the analogue of Figure 9 with claimed professional-expense deductions (*frais réels*) as the outcome. The estimation requires a new CASD export and has not been run.

E.9 Alternative sample periods

The headline specification includes all event years. The maps re-estimated excluding the 2020–2023 event years are placed here for inspection; a numerical comparison with the headline map awaits a status-stable export for the restricted period.

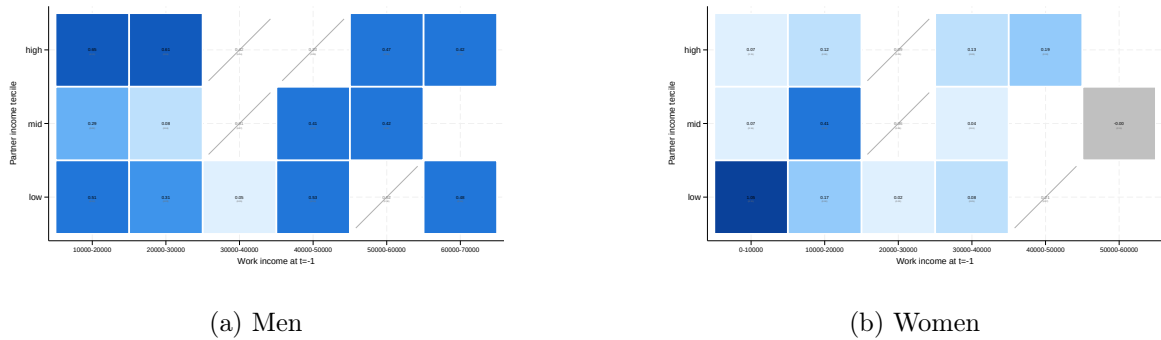


Figure E.7: Intensive-margin elasticities by own and partner income: excluding event years 2020–2023

Notes: As Figure 9, excluding the 2020–2023 event years (the extensive-margin and broad-income counterparts carry the same `_no20202023` tag in the figures folder). The maps shown predate the 22 August display conventions (the residualised inverse-variance variant described in Appendix E.8’s note, without the status-stability restriction); no status-stable export excluding 2020–2023 exists yet.

E.10 Heterogeneity

The gender breakdown of the subgroup estimates of Section 4.4, followed by placeholders for the within-subgroup maps; each will hold the maps of Figure 9 re-estimated within the indicated subgroups.

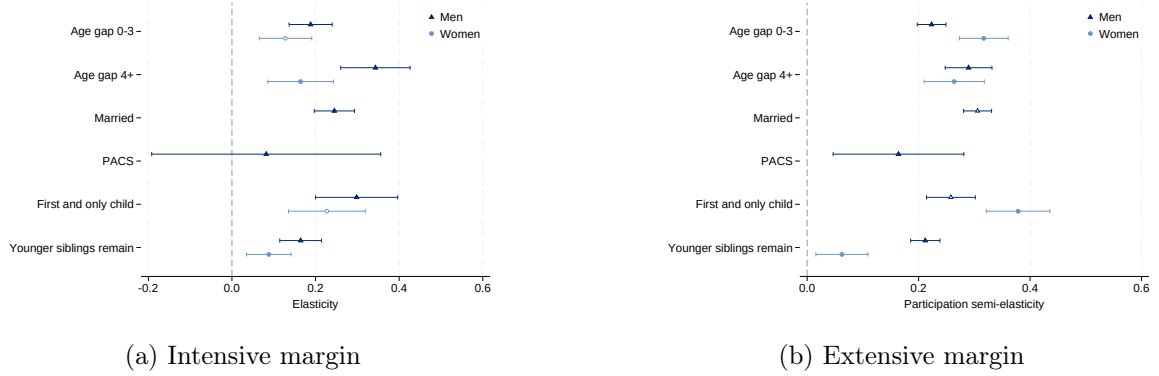
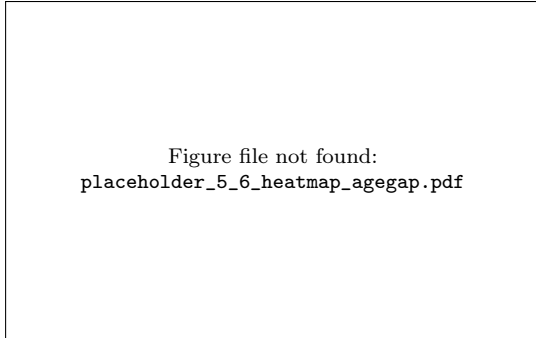
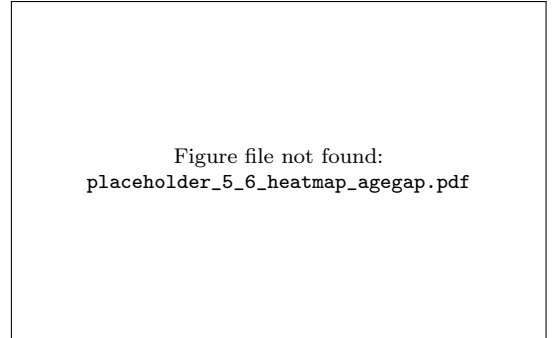


Figure E.8: Elasticities by household subgroup and gender

Notes: As Figure 10, by gender: dark triangles are men, light circles are women. Men's estimates are computed from the samestatus subgroup files; the women's series is a placeholder from the pre-status-stability exports excluding the 2020–2023 event years, and covers the age-gap and children cuts only—the married/PACS rows are men-only.



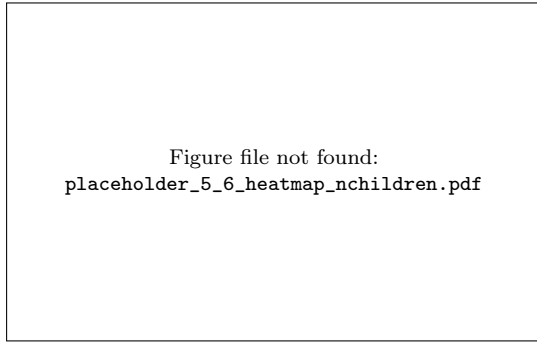
(a) Age gap 0–3



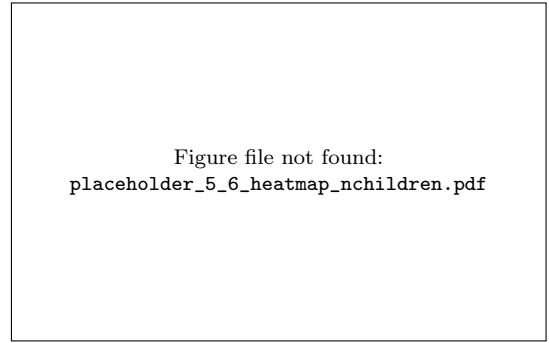
(b) Age gap 4+

Figure E.9: Elasticities by own and partner income, by age gap between the parents

Notes: Placeholder. The panels will report the intensive-margin map of Figure 9, re-estimated within households where the parents' age gap is at most three years and where it is four years or more, under the display conventions of Appendix E.1.



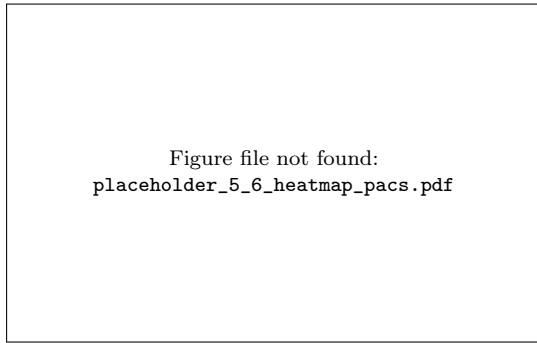
(a) First and only child



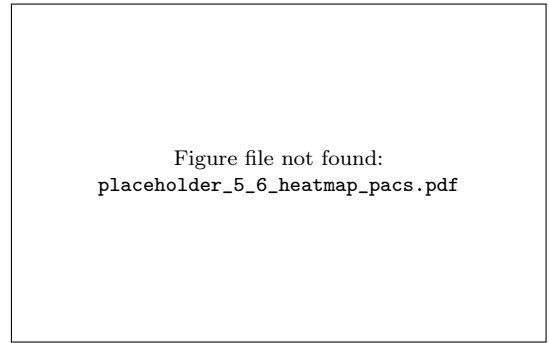
(b) Younger siblings remain

Figure E.10: Elasticities by own and partner income, by number of children

Notes: Placeholder. The panels will report the intensive-margin map of Figure 9, re-estimated within households whose exiting first-born is the only child and within households where younger siblings remain in the tax unit, under the display conventions of Appendix E.1.



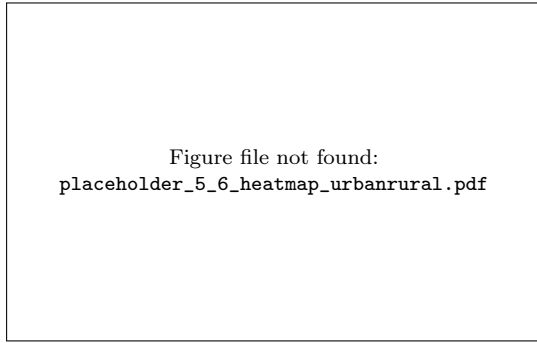
(a) Married



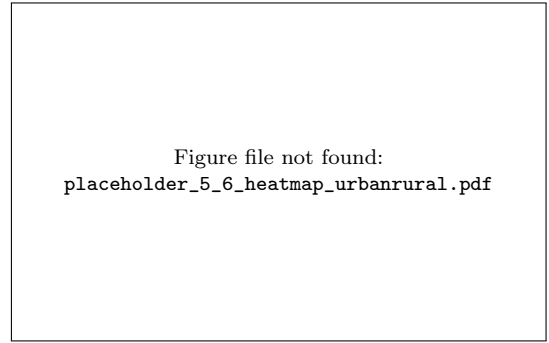
(b) PACS

Figure E.11: Elasticities by own and partner income: marriage versus PACS

Notes: Placeholder. The panels will report the intensive-margin map of Figure 9, re-estimated within married couples and within couples in a civil union (PACS), under the display conventions of Appendix E.1. Cell support is the binding constraint for the PACS panel.



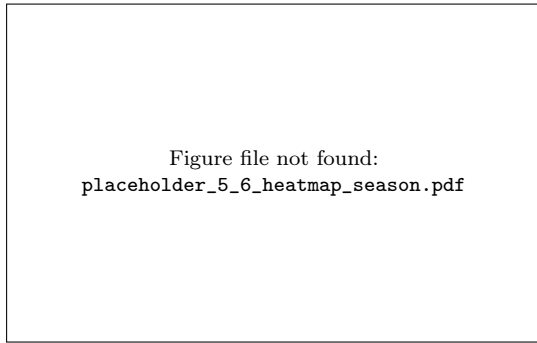
(a) Urban



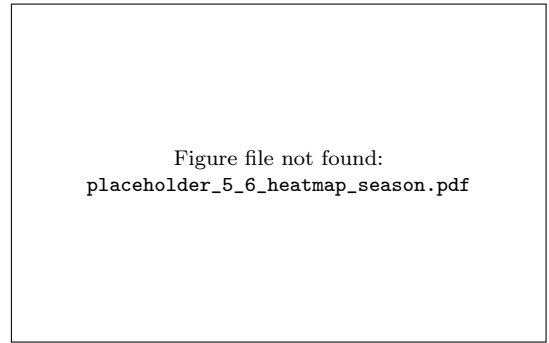
(b) Rural

Figure E.12: Elasticities by own and partner income: urban versus rural

Notes: Placeholder: no estimates are shown. The panels will report the intensive-margin map of Figure 9, re-estimated separately for households in urban and in rural municipalities of residence at $r = -1$, under the display conventions of Appendix E.1.



(a) Summer weddings



(b) Winter weddings

Figure E.13: Elasticities by own and partner income: summer versus winter marriages (before 2011)

Notes: Placeholder: no estimates are shown. This cut belongs to the marriage design of Appendix F, not to the child's fiscal exit: before the 2011 reform, the wedding date determined the first jointly taxed year, so couples marrying in summer and in winter faced the same event with different timing. The panels will report the marriage-design map for the two groups.

F Robustness: the marriage design

[This appendix will report the marriage-design counterpart of the headline results—the same negative income gradient at the marriage event, estimated on both sides of the new tax unit. The two-sided marriage exports are in preparation at CASD.]

G Monotonicity

This appendix (i) states the monotonicity condition in the canonical reform-based simulated-IV setting (Gruber & Saez, 2002; Jakobsen & Søgaaard, 2022), (ii) characterises when the condition fails and illustrates a violation in a calibrated example, and (iii) closes by arguing that violations are far less likely under schedule *shifts*, as in our design, than under reforms that change the schedule’s gradient.

G.1 The condition in the reform-based setting

Consider a reform that raises statutory rates on part of the schedule. Let $g \in \{0, 1\}$ indicate whether the reform is in force, and let $\text{MTR}_i(g)$ denote the marginal tax rate individual i faces under regime g , inclusive of behavioural responses. The simulated instrument is the mechanical prediction of $\text{MTR}_i(1) - \text{MTR}_i(0)$, holding income fixed at its pre-reform value. The IV estimand averages individual responses with weights proportional to the realised first-stage responses $\pi_i \equiv \text{MTR}_i(1) - \text{MTR}_i(0)$ (Angrist & Imbens, 1995; Borusyak et al., 2022). For the weights to be non-negative—and the estimand a convex average that cannot reverse the sign of a uniformly signed effect (Blandhol et al., 2026)—the π_i must share a common sign: first-stage monotonicity requires

$$(G.1) \quad \text{MTR}_i(1) \geq \text{MTR}_i(0) \quad \forall i,$$

with the reverse inequality for a reform that cuts rates. A violation—individuals whose realised marginal tax rate moves *against* the direction of the reform—puts negative weight on their responses, and the estimand loses its interpretation as a convex average.

G.2 When is it violated, and what does it mean?

Under the strong assumptions standard in the literature of (i) a quasi-linear utility model, implying no income effects, (ii) continuous labour supply, and (iii) no heterogeneity in elasticities, condition (G.1) is not violated. The intuition is that, for any reform—whether it increases or decreases the marginal tax rate in a given bracket—the utility ranking between the kink point and interior solutions in brackets unaffected by the reform remains unchanged.

As a result, any new interior optimum in a bracket unaffected by the reform is dominated by the kink point at the reform-affected bracket.

When either of the first two assumptions is relaxed—allowing income effects or labour-supply rigidities—and elasticities or preferences are heterogeneous, first-stage monotonicity can be violated.

We illustrate a violation of first-stage monotonicity using a model with income effects and heterogeneity in the curvature of the utility function. Instead of the standard quasi-linear utility function, we consider a more general specification that allows for income effects and non-linear labour-supply responses.⁴² Individuals derive utility from consumption c and hours worked h according to

$$u_i(c, h) = \frac{c^{1-\sigma_i}}{1-\sigma_i} - \chi_i \frac{h^{1+\varphi_i}}{1+\varphi_i},$$

where $\sigma_i > 0$ governs the curvature of utility from consumption, and hence the strength of income effects, $\varphi_i > 0$ governs the curvature of the disutility from labour, and $\chi_i > 0$ scales the marginal disutility of hours worked. Labour income is given by $y = wh$. We assume that hours rigidities restrict individuals to four discrete choices: 0, 20, 30, or 40 hours per week.

For the tax schedule, we consider the French income tax system as of 2015. In the part of the schedule on which we focus, labour income up to €26,791 per year is taxed at a marginal rate of 14%, and income above this threshold at 30%.⁴³ Because utility is defined over hours worked, we can equivalently express the tax schedule as a function of hours, with a kink at $h = k$ corresponding to the income threshold. In the example, we set the hourly wage to €20.6, so that the threshold at €26,791 corresponds to approximately 25 hours of work per week.

We study a hypothetical reform ($g = 1$) that eliminates the 30% bracket by raising the marginal tax rate above the threshold from 30% to 41%, equal to the next statutory rate in the French schedule.

We consider two preference types, Type A and Type B, and calibrate the model such that both share the same discrete optimum of 30 hours per week under no reform. The key parameter

⁴²Hours rigidities would be sufficient to illustrate a violation. To make the argument applicable more broadly, however, we consider a more general model with income effects.

⁴³To simplify the exposition, we ignore the *décote*, which does not affect the example; results including the *décote* are available on request.

governing the relative strength of income and substitution effects is the consumption-curvature parameter σ_i . Type A is calibrated with $\sigma_A < 1$, so that the substitution effect dominates, whereas Type B is calibrated with $\sigma_B \gg 1$, so that the income effect dominates. For each type, the scale χ_i is set to deliver this common optimum. The resulting preference parameters are

$$(\sigma_A, \varphi_A, \chi_A) = (0.696, 0.300, 0.272), \quad (\sigma_B, \varphi_B, \chi_B) = (7.000, 0.300, 9.17 \times 10^{-30}).$$

Under no reform, both types choose 30 hours per week, placing them in the 30% bracket:

$$\text{MTR}_A(0) = \text{MTR}_B(0) = 0.30.$$

The simulated instrument, computed by holding both types' incomes fixed at their 30-hour values, accordingly predicts the same treatment change for both: a rise in the marginal tax rate from 30% to 41%, that is, a predicted first-stage response of +0.11.

The optimal hours choices under the reform and under no reform are shown in Figure G.1.⁴⁴ Under the reform, the two types' optimal choices diverge:

$$h_A^* = 20 \text{ hours/week}, \quad h_B^* = 40 \text{ hours/week}.$$

The realised treatments therefore respond to the same shock in opposite directions:

$$\pi_A = \text{MTR}_A(1) - \text{MTR}_A(0) = 0.14 - 0.30 = -0.16 < 0, \quad \pi_B = \text{MTR}_B(1) - \text{MTR}_B(0) = 0.41 - 0.30 = +0.11$$

Type A's marginal tax rate moves against the direction of the reform: despite being exposed to a shock that raises statutory rates—and despite the instrument predicting a tax increase for it—its realised marginal tax rate *falls*, violating condition (G.1). Type B's realised treatment moves with the reform.

⁴⁴The optimal hours choice under each regime is determined by maximising $u_i(c, h)$ subject to the piecewise-linear budget constraint. There is no closed-form solution in general, so the optimum is obtained numerically by comparing utility across all candidate solutions: interior first-order-condition roots on each linear segment, the kink, and the corner at zero hours.

First-stage monotonicity is therefore violated: the estimand places negative weight on Type A's response and, with heterogeneous elasticities, can even reverse the sign of a uniformly signed response—it is not weakly causal in the sense of Blandhol et al. (2026).

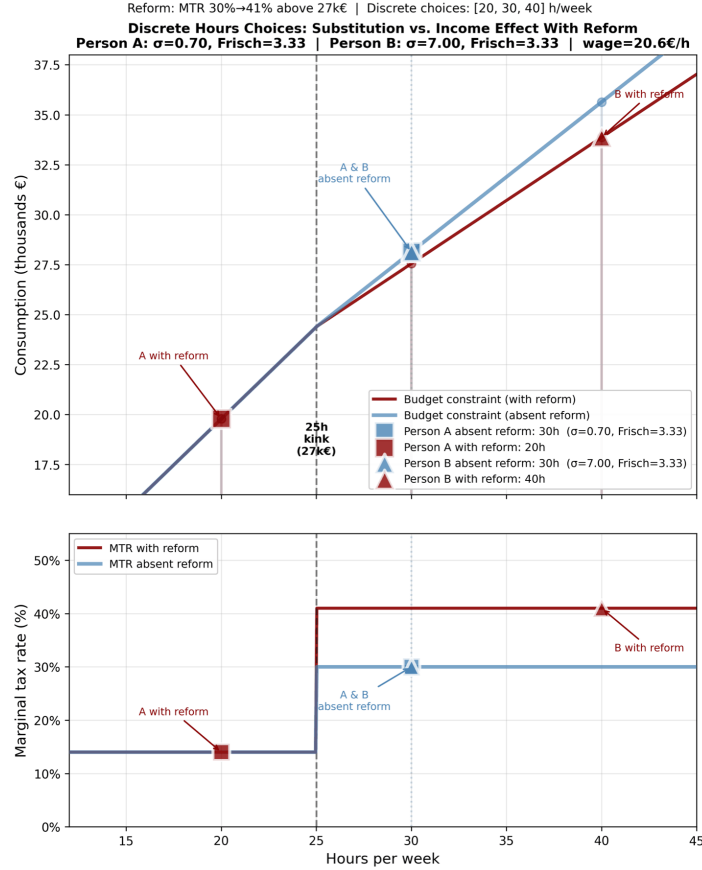


Figure G.1: Violation of first-stage monotonicity under heterogeneous labour-supply responses

Top panel. Budget constraints under the reform (red) and under no reform (blue). The reform raises the marginal tax rate above the second bracket threshold from 30% to 41%. Both Type A and Type B optimally choose 30 h/week under no reform. Under the reform, Type A chooses 20 h/week and is below the kink; Type B ($\sigma_B = 7$) chooses 40 h/week and is above the kink.

Bottom panel. Marginal tax rates under the reform (red) and under no reform (blue), as a function of hours worked, with the same optimal choices indicated.

A strong income effect for Type B is not necessary for the violation: if Type B simply stays at 30 hours under the reform ($\pi_B = +0.11$) while Type A moves to 20 ($\pi_A = -0.16$), the same violation arises.

Models with extensive-margin labour-supply responses can also generate violations of first-stage monotonicity. An extensive-margin response to, for example, a reform that increases

tax rates moves an individual out of the tax schedule altogether, that is, in France, to a marginal tax rate of zero, so that $\pi_i < 0$. If, at the same time, other individuals j remain employed and face the higher rate under the reform, $\pi_j > 0$, the realised treatment responds to the shock in opposite directions across individuals, violating condition (G.1). Because extensive-margin responses to tax increases are empirically common, as we also document in this paper, this is a realistic and quantitatively important source of monotonicity violations in reform-based IV strategies.

We stress that the reform-based setting of this example is a foil: it is the canonical design from which our identification strategy departs, not our identifying variation. We now turn to our design.

G.3 Why first-stage monotonicity is likely satisfied for our instrument

The argument below is stated for the marriage design; it carries over unchanged to the fiscal exit of a child, which changes the household tax base in the same way.

Figure G.2 illustrates, for a given level of own income, why such a monotonicity violation is highly unlikely with our instrument. The figure shows the marginal tax rate an individual faces as a function of own income, both when single (blue dashed) and when married to a partner earning €40,000 (red solid). Crucially, the marginal tax rate for a married individual remains strictly positive—at 14%—even when own income is zero, because the partner’s income already places the household above the tax-free threshold: even the most extreme reduction in labour supply, a complete exit, leaves the individual facing a 14% rate rather than a 0% rate. Since the instrument in our case study captures individuals predicted to move from the 14% to the 30% bracket, a violation would require the realised marginal rate to fall strictly below 14% with marriage—impossible unless partner income collapses dramatically. The logic extends to all individuals facing predicted tax increases; for predicted decreases, violations are likewise ruled out, since they would require a dramatic increase in labour supply.

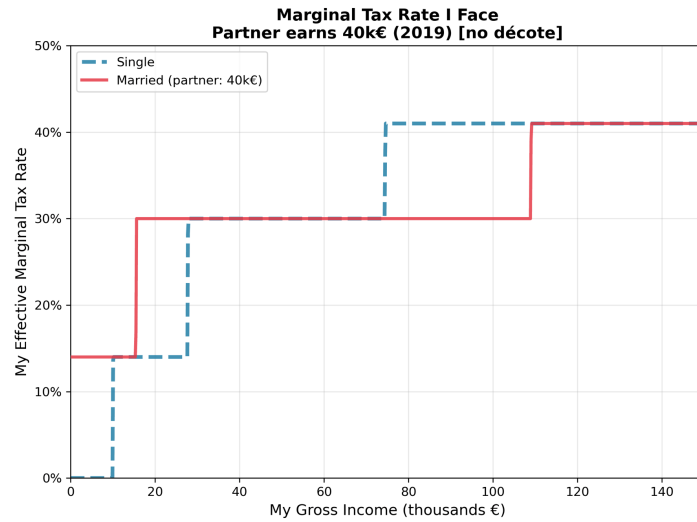


Figure G.2: Marginal tax rate under marriage, partner earns €40,000

Notes: The figure shows the effective marginal tax rate as a function of own gross income under individual taxation (blue dashed) and under joint taxation when married to a partner earning €40,000 (red solid), based on the 2019 French income tax schedule without the *décote*. Income is not pooled, so own consumption corresponds to own after-tax income only. As a result, the marginal tax rate for a married individual is 14% even at zero own income, because the partner's income of €40,000 already places the household above the tax-free threshold. Consequently, even a complete exit from the labour market cannot lower the individual's marginal tax rate below 14%, ruling out a monotonicity violation for individuals predicted to move from the 14% to the 30% bracket.

H Evidence underlying the literature map

Appendix Table H.1 records the estimates and distributional locations underlying Figure 1, together with each estimate’s design, estimand, horizon and population; ranking universes and source versions are documented in the notes. It includes only studies represented in the figure, while retaining alternative estimates from those studies for context.

Table H.1: Evidence base for the literature map of work income responses

Study	Outcome	Design	Estimand	Horizon	Population	Estimate (SE)	Identifying variation
Saez (2010)	Work income	Bunching	Compensated	1995-2004 pooled	EITC filers with one child, wage earners only	0.025 (0.036)^a	P13.4–P15.3 (C)
					EITC filers with two or more children, wage earners only	0.003 (0.022)^a	P19.1–P21.1 (C)
	Earnings				All EITC filers with one child	0.213 (0.033)	P13.4–P15.3 (C)
					Self-employed EITC filers with one child	1.101 (0.092)	
					All EITC filers with two or more children	0.152 (0.02)	P19.1–P21.1 (C)
Chetty, Friedman and Saez (2013)	Work income	Reform DiD (EITC around first-child birth)	Frictionless-model elasticity matching observed EITC refund changes	Event study around first-child birth, 2000-2005	EITC-eligible parents around the birth of the first child, 2000-2005	0.755 (0.054)	
						0.31 (0.018)^b	P0–P19.4 (C)
						0.14 (0.015)^b	P24.8–P54.5 (C)
Chetty et al. (2011)	Work income	Reform IV (small reforms)	Observed elasticity; the paper argues frictions attenuate it	1994-2001 pooled	All wage earners	0.19 (0.019)	P0–P54.5 (C)
	Work income (taxable base)	Bunching	Compensated (observed, frictions not netted out)			0.21 (0.012)	
	Taxable income	Structural calibration	Structural lower bound (calibrated, frictions ≤ 5% of consumption)		Wage earners	-0.005 (0.003)	not locatable
	Work income (taxable base)	Bunching	Compensated (observed)		Married women, wage earners	~0.01 ^j	P75 (A)
Bastani and Selin (2014)	Work income (taxable base)	Bunching	Compensated	1999-2005 pooled	Wage earners aged 15-70; anyone with active business income or closely-held-corporation ties removed	≥0.34	not locatable
			Structural upper bound (tolerated loss = 1% of disposable income)		Wage earners aged 15-70	~0.02	P75 (A)
	Taxable income		Compensated	2000-2008 pooled	All self-employed	~0.00 ^k	P80 (A)
Lehmann et al. (2013)	Work income incl. contributions	Reform IV	Marginal-rate elasticity, average net-of-tax rates controlled	1-year, 2003-2006 pooled	Private-sector employees earning less than twice the minimum wage; public sector and self-employed excluded	~0.00	
						0.024	
Saez (2003)	Work income	Reform IV (bracket creep, 1979-1981 inflation)	Compensated	1-year, 1979-1981	Tax returns in the NBER panel, 1979-1981; singles with AGI \$3,000-40,000 and married \$5,000-70,000 (1979 USD)	0.225 (0.085)^g	P10–P60 (C)
						0.016 (0.096)	
Kleven and Schultz (2014)	Earnings	Reform IV	No income effects	3-year, 1984-2005	All workers, top 10 percent earners	0.084 (0.173)^f	P44–P94 (C)
						0.120 (0.196)	P55–P94 (C)
					All workers, top 20 percent earners	0.085 (0.012)^c	P90–P100 (A)
						0.076 (0.008)^c	P80–P100 (A)

Continued on next page

Table H.1 continued

Study	Outcome	Design	Estimand	Horizon	Population	Estimate (SE)	Identifying variation
Jakobsen and Sogaard (2022)	Work income	Local reform IV (2SLS local linear)	No income effects	4-year, 2008-2012 (pre-period 2004-2008)	Wage earners, top 10 percent earners	0.081 (0.012)	P90–P100 (A)
					Wage earners, top 20 percent earners	0.073 (0.009)	P80–P100 (A)
	Work income incl. contributions plus transfers				Aged 15+, pre-reform personal income above DKK 250k; self-employed excluded	~ 0.038	P74.9–P76.9 (C)
						~ 0.051	P82.5–P83.8 (C)
						~ 0.047	P87.7–P88.5 (C)
						~ 0.031	P91.1–P91.5 (C)
						~ -0.004	P93.3–P93.6 (C)
						~ 0	P94.9–P95 (C)
						~ 0.028	P95.6–P95.7 (C)
						~ -0.001	P96.2–P96.3 (C)
Miao, Selin and Soderstrom (2025)	Taxable income	Reform IV		4-year, 2008-2012		0.016 (0.007)	P74.9–P100 (C)
	Earnings	Reform difference-in-differences (grouped)	No income effects (grouped Wald / DiD)	4-year, 2015 vs 2019	Percentile groups 96-100 (treatment) vs 88-95 (control), aged 20-65	0.156 (0.023)	P96–P100 (A)
					Percentile groups 96-100 vs 88-95, groups 94-97 excluded	0.171 (0.019)	
Kleven et al. (2025)	Work income incl. contributions	Reform IV, job switchers	Long-run population elasticity under orthogonality of switching timing	Four-year	Treated job switchers above the threshold	0.487 (0.026)^d	P70–P98 (A)
Gelber (2014)	Work income	Reform IV	Uncompensated	Two-year	Treated job switchers in the top decile	0.642 (0.029)^d	P90–P98 (A)
					All treated job switchers	0.486 (0.022)	P70–P98 (A)
					Treated job switchers in the top decile	0.725 (0.029)	P90–P98 (A)
					Married men aged 18-65 with positive earnings in 1989 or 1990	0.210 (0.02)^e	not locatable
					Married women aged 18-65 with positive earnings in 1989 or 1990	0.230 (0.03)^e	
					Married men aged 18-65 with positive earnings in 1989 or 1990	0.410 (0.02)	
					Married women aged 18-65 with positive earnings in 1989 or 1990	0.470 (0.02)	
Sigaard (2023)	Work income	Reform IV	No income effects	3-year	Danish wage earners with reported contract and performed hours, 2002-2016	0.058 (0.0037)	not locatable
Blomquist and Selin (2010)	Hours worked	Reform IV	Uncompensated	10-year, 1981 vs 1991		0.082 (0.0091)	
	Taxable income					0.095 (0.0042)	
	Work income (taxable base)				Married or cohabiting men; self-employed excluded	0.210 (0.074)^l	not locatable
He et al. (2021)	Work income (taxable base)				Married or cohabiting women; self-employed excluded	1.442 (0.832)^l	
		Reform IV	Compensated (income effects small and insignificant)	One-year	Wage earners aged 18-60 in one (unnamed) Chinese city	2.423 (0.188)^m	not locatable
		Bunching		Pooled months		0.1 (0.02)	
						0.41 (0.07)	
Kleven–Waseem (2013)	Work income (taxable base)	Reform IV (revised, long-run)		Long-run		0.09 (0.03)	
						~4	
		Bunching (notch)	Observed (friction-attenuated) elasticity	Annual, 2006-2007	Pakistani wage-earner filers 2006-07 (under 2% of the working-age population, over 99% male)	~ 0 (0.000-0.004 across notches)ⁿ	not locatable

Continued on next page

Table H.1 continued

Study	Outcome	Design	Estimand	Horizon	Population	Estimate (SE)	Identifying variation
			Structural (frictionless) elasticity, bunching-hole lower bound			0.024 (0.009)	
						0.035 (0.010)	
						0.034 (0.019)	
						0.001 (0.013)	

Notes: One row per coded estimate; **bold** estimates are plotted in Figure 1, with superscripts matching the figure’s label letters. Within a study block, repeated entries are shown once. (–) indicates that no standard error is reported.

Placement. Ranks in the identifying-variation column: (A) reported by the study; (B) computed by us from a closely matching official distribution; (C) computed from an official distribution with a related income concept, drawn dashed in the figure; (D) weak reconstruction, (Q) qualitative statement, (U) unrecoverable — not placed, shown in the figure’s right panel. The column gives the income range of the treated group only, never the estimation sample: control groups located elsewhere in the income distribution (Miao et al.: P88–95; Kleven et al.: below the reform threshold; Jakobsen–Søgaard: the validation region) are not part of the stated range, while for Kleven–Schultz the sample and the treated band coincide because identification uses within-sample variation in reform exposure.

Estimands. Not harmonised: bunching estimates are compensated elasticities; Gelber’s plotted pair is uncompensated (the study’s compensated estimates are 0.41 and 0.47); most register-based reform estimates are specified without income effects; and Chetty–Friedman–Saez report the elasticity a frictionless model requires to match observed EITC refund changes. Reported income effects are small throughout, so these differences are second order relative to design and horizon. Statutory-taxable-income estimates on wage-earner samples enter as work income (taxable base); population-wide taxable-income studies remain in the project archive.

Sources. Values are taken from the published articles, with exceptions: Chetty et al. (2011) from the revised NBER working paper; Sigurdsson (2025) from the November 2024 manuscript; Blomquist–Selin verified from CESifo Working Paper 2644. Jakobsen–Søgaard points and confidence bands are digitized from their Figure 3, Panel D; their local estimates use triangular weights within DKK 50,000 of each centre. Kleven–Waseem’s observed elasticity is derived as the paper instructs (structural lower bound times the responding share); no standard error applies.